On the unitary dual of $\overline{SL(2,F)}$, F p-adic

Draft 3. Corrected and expanded version

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Abstract

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Let $\widetilde{G}$ be Kubota's n-fold cover of $\mathrm{SL}(2,F)$, where F is a p-adic field containing the nth roots of unity. The preimage $\widetilde{A}$ of the diagonal torus is not abelian when $n>2$. We describe its genuine irreducible representations by a Stone–von Neumann argument. We prove that normalized split orbital integrals of genuine test functions vanish off the center of $\widetilde{A}$, and we determine their image when $\gcd(p,2n)=1$. We compute the trace and the character of a genuine principal series representation. The character involves Weyl conjugation in the cover, not inversion. When $\gcd(p,2n)=1$ and n is odd or divisible by 4, we deduce a character identity between genuine principal series of $\widetilde{G}$ and principal series of $\mathrm{SL}(2,F)$ for test functions whose split orbital integrals match. The case $n\equiv 2 \pmod 4$, which includes the double cover, is excluded from this identity. These are the paper's contributions. We do not classify the unitary dual. Instead we compare each unitary-dual claim of the 1999 manuscript and of the author's 2001 paper with what is proved here, and we survey the literature on the rest.

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1 Introduction

Kubota's cover of $\mathrm{SL}(2,F)$ has a nonabelian torus, and that one fact shapes everything below. We state the problem and the contributions, compare them with the 1999 and 2001 accounts of the unitary dual, and fix the hypotheses.

Let F be a nonarchimedean local field of characteristic zero, with residue characteristic p. Fix $n \geq 1$ and assume that F contains the group μ_n of nth roots of unity, of order n. Kubota [Ku67] constructed a central extension

$$1 \longrightarrow \mu_n \longrightarrow \widetilde{G} \xrightarrow{\rho} G = \mathrm{SL}(2,F) \longrightarrow 1.$$

The 1999 title writes $\overline{SL(2,F)}$ for $\widetilde{G}$. A representation of $\widetilde{G}$ is *genuine* if μ_n acts on it through a fixed faithful character. Genuine representations are the local objects of the theory of metaplectic forms of Kubota, Gelbart, Flicker and Kazhdan–Patterson (Section 2).

The 1999 manuscript had two aims. The first was to describe the genuine unitary dual of $\widetilde{G}$. The second was to relate genuine principal series of $\widetilde{G}$ to principal series of G by a character identity. The motivation was a planned proof of Labesse's multiplicity one conjecture for $\mathrm{SL}(2)$ over a number field, which needed the case $n = 2$.

A basic difficulty is that the preimage $\widetilde{A}$ of the diagonal torus is not abelian when $n > 2$. Its genuine irreducible representations have dimension $d = n/\gcd(n,2)$, and their characters vanish off the center of $\widetilde{A}$. For example, when $n = 3$ every genuine irreducible representation of $\widetilde{A}$ has dimension three. This affects both the orbital integrals and the characters of induced representations.

Contributions. The paper makes three contributions.

(A) An explicit treatment of the covering torus: its center, its genuine irreducible representations and their traces, and the action of the Weyl element (Section 5).

(B) A normalized image theorem for split orbital integrals of genuine test functions (Section 6).

(C) A restricted split-torus trace identity between genuine principal series of $\widetilde{G}$ and principal series of $\mathrm{SL}(2,F)$ (Section 9), built on a direct computation of the trace and character of genuine principal series (Section 8).

Main results. In detail, the paper proves the following.

(i) Theorem 5.6 describes the genuine irreducible representations of $\widetilde{A}$ by their central characters and computes their traces.

(ii) Lemma 6.1 and Theorem 6.4 show that normalized split orbital integrals vanish off the center Z_A of $\widetilde{A}$, and identify their image.

(iii) Lemma 8.1, Corollary 8.2 and Theorem 8.3 compute the trace and the character of a genuine principal series representation.

(iv) Lemma 7.3 shows that an unramified genuine principal series has a one-dimensional space of spherical vectors.

(v) Theorem 9.4 gives the character identity

$$\mathrm{tr}\,\mathrm{I}(\chi_\lambda)(\bar{f}) = \mathrm{tr}\,\mathrm{Ind}_B^G(\eta_\lambda \delta_B^{1/2})(f)$$

for test functions $\bar{f}$ on $\widetilde{G}$ and f on G whose split orbital integrals match.

Along the way it proves Kubota's explicit splitting of $\mathrm{SL}(2,\mathcal{O})$ (Lemma 3.3) and the rank-one Jacquet-module computation (Lemma 7.1), so that these results do not rest on references with other hypotheses.

How the results depend on each other.

Lemma 3.2 (cocycle)	$\longrightarrow$ Lemma 5.2 (center)	$\longrightarrow$ Theorem 5.6 (torus traces)
Lemma 3.3 and (3.4)	$\longrightarrow$ Lemma 6.3 (constant term)	$\longrightarrow$ Theorem 6.4 (image)
Theorem 5.6 and Lemma 6.3	$\longrightarrow$ Lemma 8.1 (trace)	$\longrightarrow$ Theorem 8.3 (character)
Corollary 8.2 and Lemma 9.1	$\longrightarrow$ Theorem 6.4 (existence)	$\longrightarrow$ Theorem 9.4 (matching)

Relation to the 1999 and 2001 unitary-dual accounts. The 1999 manuscript described the genuine unitary dual by reducing it, through Jacquet's theorem, to principal series, complementary series, the representations at the end of the complementary interval, and supercuspidal representations. The author's 2001 paper [Jo01] restates these claims in its Section 3 and adds a list of the types of unitary and tempered representations, for $\gcd(p,2n)=1$. That list is only as reliable as the claims it rests on. Table 1 compares each claim with what is proved here. Here s_{old} is the 1999 parameter, and Section 7.3 converts it to the parameter s of this paper.

Claim of the 1999 manuscript (repeated in [Jo01])	Status in this draft
(a) Genuine representations of $\widetilde{A}$ come 2-to-1 from characters of $h(\varpi^{n\mathbb{Z}}\mathcal{O}^{\times})$.	False as stated. Corrected and proved: Theorem 5.6.
(b) Split orbital integrals of genuine functions fill out the Weyl-invariant genuine functions on the center.	Proved: Theorem 6.4, with the normalizations compared in Remark 6.5.
(c) Unitary principal series: irreducible for n even, and for n odd reducible exactly when $\chi^{\tilde{w}}=\chi\neq 1$ (attributed to Moen).	Irreducibility for Weyl-regular data is proved (Proposition 7.2). The reducible cases are source checked against [GS16] (Remark 2026.2).
(d) $\mathrm{I}(\chi)$ is a complementary series for $0<s_{\text{old}}<1/n$, by extending it to a cover of $\mathrm{GL}(2,F)$ and applying [FK86, §§26–27].	Not proved. The extension step was never proved (Remark 2026.3).
(e) At $s_{\text{old}}=1/n$: a square-integrable subrepresentation and a non-tempered spherical quotient, unitary by [KP84, Thm. II.2.1].	Cited only (Remark 2026.3).
(f) The list of all unitary and tempered types (in [Jo01]).	Cited only. It follows from (c)–(e) and Jacquet's theorem (Remark 2026.4).
(g) The principal-series character uses $\mathrm{tr}\,\chi(a)+\mathrm{tr}(\chi(a)^{-1})$.	False for $n>2$. Corrected and proved with Weyl conjugation: Theorem 8.3.
(h) The character identity holds for every n with $\gcd(p,2n)=1$.	Proved only for n odd or $4\mid n$: Theorem 9.4.

Table 1: The 1999 and 2001 claims and their status in this draft.

So this draft proves (a), (b), (g) and (h) in corrected form, and the Weyl-regular part of (c). It

neither proves nor refutes (d), (e) and (f). The title is the 1999 title, kept by the conventions of this series of revisions. A more accurate description of the present content would be "the covering torus and split principal-series characters for Kubota covers of $\mathrm{SL}(2)$".

Restriction on n. Theorem 9.4 assumes $\gcd(p,2n)=1$ and that n is odd or divisible by 4. It does not cover $n \equiv 2 \pmod 4$, and in particular it does not cover the double cover $n=2$ that motivated the 1999 manuscript. Remark 9.6 explains why.

Status labels. Every result is of one of four kinds. *Proved here* results are theorems, propositions and lemmas with proofs. The others are stated in remarks numbered "Remark 2026.n", each of which opens with its status. *Source checked* means that the statement was compared with the cited source, at least with its abstract and introduction. *Cited only* means that the source was not consulted for this revision. *Conjectural* means that no proof is known to us. Open problems are collected in Section 10.

Standing hypotheses. Throughout, $|\mu_n(F)| = n$. The description of the center in Lemma 5.2, and Lemma 6.1, need nothing more. From Section 5.2 on we assume $\gcd(p,n)=1$. When the splitting of $\mathrm{SL}(2,\mathcal{O})$ is used, that is, in the image theorem and in all trace computations, we assume $\gcd(p,2n)=1$, as in the 1999 manuscript. The splitting itself needs only $\gcd(p,n)=1$ (Lemma 3.3). We do not assume $\mu_{2n} \subset F$. Each theorem restates its hypotheses.

Outline. Section 2 reviews the literature. Section 3 explains the background, and Section 4 collects the notation. Sections 5–9 contain the results. Section 10 summarizes them and lists open questions. A glossary follows.

2 Literature review

The relevant literature divides at 1999, the date of the original manuscript. A result for a cover of $\mathrm{GL}(2,F)$ does not automatically give the analogous result for the cover of $\mathrm{SL}(2,F)$, so we say which group each work treats.

2.1 Work before 1999

The cover and the double cover. Kubota [Ku67] constructed the n-fold covering group of $\mathrm{SL}(2,F)$ by an explicit two-cocycle built from the nth Hilbert symbol. This cocycle is the starting point of Sections 3 and 5. Gelbart [Ge76] developed the representation theory and the spectral theory of the double cover, with the Weil representation as the central example. Gelbart and Sally [GS75] studied the intertwining operators of the double cover and the decomposition of the Weil representation. Their work explains why the exceptional constituents of induced representations must be treated separately from the generic principal series.

Principal series of covers of SL(2)**.** Moen [Mo] studied intertwining operators for covering groups of SL(2) over a local field. Moen [Mo88] proved the irreducibility of certain unitary principal series of these covers when p is odd. The 1999 manuscript attributes to Moen the statement that for $\gcd(p,n)=1$ all unitary genuine principal series are irreducible when n is even, and that for n odd they are reducible exactly when $\chi^{\tilde{w}}=\chi\neq 1$. Goldberg and Szpruch later proved a statement of this kind (see below). Aritürk [Ar80] determined composition series of principal series of the threefold cover. For example, he showed that off the line $\mathrm{Re}\, s=0$ the genuine principal series of the threefold cover are irreducible except at $s=\pm 1/3$. The present paper does not reprove these results. Section 7 cites them.

Parabolic induction. Bernstein and Zelevinsky [BZ76, BZ77] developed parabolic induction and Jacquet modules for reductive p-adic groups. Bernstein's notes [BD84] remark that these arguments carry over to finite central extensions. This is the basis of the Jacquet-module statements in Section 7.

Covers of GL **and the metaplectic correspondence.** Flicker [Fl80] studied automorphic forms on covers of GL(2) and compared them with forms on GL(2) by matching orbital integrals. Kazhdan and Patterson [KP84] developed metaplectic forms on covers of $\mathrm{GL}(r)$, including the principal series and the exceptional representations. Flicker and Kazhdan [FK86] proved the metaplectic correspondence between genuine representations of covers of $\mathrm{GL}(r)$ and representations of $\mathrm{GL}(r)$. Their comparison of orbital integrals is the model for Section 9. They do not, by themselves, give a transfer for the cover of SL(2).

Supercuspidal representations and the author's related work. Manderscheid [Ma84a, Ma84b] described the supercuspidal representations of SL(2) and its double cover. Joyner [Jo88] studied a metaplectic analogue of Kazhdan's endoscopic lifting. Joyner [Jo98] proved a correspondence between generalized Hecke algebras of $\mathrm{SL}(2,F)$ and of its n-fold cover. It gives a correspondence of representations of sufficiently large level, not a classification of all of them. Schultz [Sc98] studied the lifting of characters between the double cover of $\mathrm{SL}(2,F)$ and $SO_{1,2}(F)$. That is a p-adic analogue of Adams' lifting for real groups, and it concerns the case $n=2$ excluded from Theorem 9.4.

2.2 Work from 1999 on

The author's papers. Joyner [Jo99] described the supercuspidal representations of the n-fold cover when $\gcd(p,2n)=1$, following Manderscheid. That paper proves a Frobenius formula for their characters and a character relation with $\mathrm{SL}(2,F)$ when $n=2$. Joyner [Jo01] studied invariant distributions on covers of $\mathrm{GL}(r,F)$ and the unitary and tempered dual of the cover of $\mathrm{SL}(2,F)$ for $\gcd(p,2n)=1$. It cites the 1999 manuscript as a preprint. Its Section 3 restates the unitary-dual claims of the 1999 manuscript and lists the resulting types of representations. Table 1 gives the status of each claim. It also states the image theorem for split orbital integrals in a different normalization (Remark 6.5).

General covering groups. McNamara [Mc12] developed the principal series of central extensions of split reductive groups. He uses a Stone–von Neumann theorem for the covering torus, computes the Jacquet modules of principal series, and proves a Satake isomorphism. He assumes that F contains the $2n$th roots of unity, that is, $2n \mid q-1$, and that $p \nmid 2n$. That is stronger than the hypothesis $\mu_n \subset F$ used here. For example, $F = \mathbb{Q}_5$ with $n = 4$ is allowed here but not in [Mc12]. So we use [Mc12] only for comparison. Section 5 is the rank-one analogue of his torus theory, and Lemma 7.1 is proved directly. Li [Li12] developed local harmonic analysis for finite central covers of connected reductive groups. That work includes a Plancherel theorem [Li12, Thm. 2.6.4], a Paley–Wiener theorem [Li12, Thm. 5.6.3] and Kazhdan's density theorem [Li12, Thm. 5.8.10]. Tang [Ta17] computed Plancherel measures of principal series of central extensions of split groups by a comparison method. Tang also constructed genuine central characters of the metaplectic torus in the simply laced case. Luo [Lu20] proved the Knapp–Stein dimension theorem for finite central covers.

The cover of $\mathrm{SL}(2)$. Goldberg and Szpruch [GS16] computed the Plancherel measures of genuine principal series of the n-fold cover of $\mathrm{SL}(2,F)$ when $\mu_n \subset F$ and $p \nmid n$. As an application they found all reducible genuine principal series induced from unitary data [GS16, Thm. 5.2]. Such a representation is reducible if and only if n is odd and the projection of its central character to the dth powers is a nontrivial quadratic character. Szpruch [Sz19] defined a matrix of local coefficients for these representations. He does not assume that $p \nmid n$, but he assumes $4 \nmid n$. Szpruch [Sz25] computed the Whittaker spaces of the two summands of a reducible unitary genuine principal series, without assuming $p \nmid n$.

How this paper fits. Sections 5, 6, 8 and 9 are elementary and self-contained, apart from the Weyl integration formula and standard facts about the cover. Section 7 relies on the works cited above, and its statements should be read with the hypotheses of those works.

3 Background

Everything in the paper rests on an explicit cocycle, explicit Haar measures and the classical theory for $\mathrm{SL}(2,F)$. We cover the Hilbert symbol, Kubota's cover and its splitting over $\mathrm{SL}(2,\mathscr{O})$, genuine representations and functions, Haar measures, and the orbital integrals and principal series of the linear group. Each subsection ends with an example.

3.1 The Hilbert symbol

The cover is built from the nth Hilbert symbol. We recall the properties that we use.

Let $\mathscr{O}$ be the ring of integers of F, ϖ a uniformizer and q the order of the residue field. Write $|\cdot|$ for the absolute value with $|\varpi| = q^{-1}$, and v for the valuation. The nth Hilbert symbol is a map $(\ ,\ )_n : F^\times \times F^\times \to \mu_n$. We identify μ_n with a subgroup of $\mathbb{C}^\times$ by a fixed isomorphism. The symbol has the following properties [Ne86].

(H1) It is multiplicative in each variable.

(H2) It is antisymmetric: $(a,b)_n(b,a)_n = 1$.

(H3) $(a,-a)_n = 1$. Hence $(a,a)_n = (a,-1)_n$.

(H4) It is nondegenerate: $(a,b)_n = 1$ for all b if and only if $a \in (F^\times)^n$.

(H5) (Tame formula.) If $p \nmid n$, write $\alpha = v(a)$ and $\beta = v(b)$. Then $(a,b)_n$ is the image in μ_n of

$$\left((-1)^{\alpha\beta} a^\beta b^{-\alpha} \bmod \varpi\right)^{(q-1)/n}.$$

In particular $(u,u')_n = 1$ for units u,u'.

Some authors use the inverse of this symbol. Every statement below is unchanged by that choice.

By (H3), $(t,-1)_n^2 = (t,1)_n = 1$, so $(t,-1)_n = \pm 1$. If n is odd, then $(t,-1)_n = (t,(-1)^n)_n = (t,-1)_n^n = 1$.

Example 3.1. Take $F = \mathbb{Q}_7$ and $n = 3$. Then $q = 7$ and $\mu_3 \subset \mathbb{Q}_7$. By (H5), $(7,3)_3$ is the image of $(3^{-1})^2 \equiv 5^2 \equiv 4 \pmod 7$. Since $4^3 \equiv 1$ and $4 \neq 1$, $(7,3)_3$ is a nontrivial cube root of unity. On the other hand $(2,3)_3 = 1$, since 2 and 3 are units.

Take $F = \mathbb{Q}_5$ and $n = 4$. By (H5), $(5,-1)_4$ is the image of $(-1)^{-1} \equiv -1 \pmod 5$, raised to the power $(5-1)/4 = 1$. So $(5,-1)_4 = -1$. This value is used in Example 5.5.

3.2 Kubota's cover

Kubota's cocycle is explicit enough to compute products of lifts of diagonal, unipotent and Weyl elements, and to write down a splitting over $\mathrm{SL}(2,\mathscr{O})$.

Let $G = \mathrm{SL}(2,F)$ and put

$$h(t) = \begin{pmatrix} t & 0 \\ 0 & t^{-1} \end{pmatrix}, \qquad u(x) = \begin{pmatrix} 1 & x \\ 0 & 1 \end{pmatrix}, \qquad w = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}.$$

Let $A = h(F^\times)$, $N = u(F)$, $B = AN$ and $W = N_G(A)/A = \{1,w\}$. For $g = \left(\begin{smallmatrix} a & b \\ c & d \end{smallmatrix}\right)$ put $x(g) = c$ if $c \neq 0$ and $x(g) = d$ if $c = 0$. Kubota's cocycle is

$$\beta(g_1,g_2) = \left(\frac{x(g_1g_2)}{x(g_1)}, \frac{x(g_1g_2)}{x(g_2)} \right)_n. \tag{3.1}$$

It is a two-cocycle [Ku67, §1]. The group $\widetilde{G}$ is the set $G \times \mu_n$ with the product $(g,\zeta)(g',\zeta') = (gg', \zeta\zeta'\beta(g,g'))$, and ρ is the projection to G. For $H \subset G$ write $\widetilde{H} = \rho^{-1}(H)$. Put $\widetilde{A} = \widetilde{A}$, $\widetilde{B} = \widetilde{B}$,

$$\tilde{h}(t) = (h(t),1), \qquad \tilde{w} = (w,1).$$

We identify $\zeta \in \mu_n$ with $(1,\zeta) \in \widetilde{G}$, which is central.

Lemma 3.2. *(a)* $\beta(h(a),h(b)) = (b,a)_n$.

(b) $\beta(u(x),u(y)) = 1$ *and* $\beta(h(t),u(x)) = 1$. *Hence* $u(x) \mapsto (u(x),1)$ *is a splitting of* N, *and* $\tilde{h}(t)(u(x),1) = (h(t)u(x),1)$.

(c) The splitting of N is unique, and $\widetilde{A}$ normalizes its image.

(d) The preimage of $\{\pm I\}$ is central in $\widetilde{G}$.

Proof. (a) We have $x(h(a)) = a^{-1}$, $x(h(b)) = b^{-1}$ and $x(h(ab)) = (ab)^{-1}$. So $\beta = (b^{-1}, a^{-1})_n = (b,a)_n$ by (H1).

(b) Unipotent matrices have $x = 1$, so $\beta(u(x), u(y)) = (1,1)_n = 1$. Next, $h(t)u(x)$ has lower row $(0, t^{-1})$, so $x(h(t)u(x)) = t^{-1} = x(h(t))$ and $x(u(x)) = 1$. Thus $\beta(h(t), u(x)) = (1, t^{-1})_n = 1$.

(c) Two splittings of N differ by a homomorphism $F \to \mu_n$. Every element of F is n times another, since F has characteristic zero. So such a homomorphism ϕ satisfies $\phi(x) = \phi(x/n)^n = 1$. Conjugation by an element of $\widetilde{A}$ maps a splitting of N to a splitting of N, because A normalizes N. By uniqueness it preserves the image.

(d) For $\varepsilon = \pm 1$ and a lift z of εI, the commutator $g \mapsto z\tilde{g}z^{-1}\tilde{g}^{-1}$ depends only on $g = \rho(\tilde{g})$. It is a homomorphism $G \to \mu_n$, because z is central modulo μ_n. The group $\mathrm{SL}(2,F)$ is perfect, so the homomorphism is trivial. □

Let $K = \mathrm{SL}(2, \mathscr{O})$. For $m \geq 1$ let K_m be the kernel of reduction modulo ϖ^m.

The splitting of K. Kubota [Ku67] showed that $\widetilde{G}$ splits over K when $p \nmid n$. We use the explicit splitting $s(k) = (k, \kappa(k))$, where

$$\kappa\begin{pmatrix} a & b \\ c & d \end{pmatrix} = \begin{cases} (c,d)_n, & 0 < |c| < 1, \\ 1, & \text{otherwise,} \end{cases} \tag{3.2}$$

and we prove that it is a splitting. For a unit u let $\omega(u) \in \mu_n$ be the image of $\bar{u}^{(q-1)/n}$, where $\bar{u}$ is the residue of u. Then (H5) reads $(a,b)_n = \omega\big((-1)^{\alpha\beta} a^\beta b^{-\alpha}\big)$, with $\alpha = v(a)$ and $\beta = v(b)$. Note that ω is multiplicative and that $\omega(u) = 1$ when $u \equiv 1 \pmod{\varpi}$.

Lemma 3.3 (Kubota's splitting)**.** *Assume* $\gcd(p,n) = 1$. *For all $k_1, k_2 \in K$,*

$$\kappa(k_1 k_2) = \kappa(k_1)\kappa(k_2)\beta(k_1, k_2).$$

Hence $s : K \to \widetilde{G}$ is a homomorphism. Moreover $\kappa = 1$ on upper triangular matrices, so $s(b) = (b,1)$ for $b \in B \cap K$.

Proof. Generators. The group K is generated by the matrices $u(x)$ with $x \in \mathscr{O}$ and by w. Indeed, let $k = \left(\begin{smallmatrix} a & b \\ c & d \end{smallmatrix}\right) \in K$. If c is a unit, then $k = u\big((a-1)/c\big)\,\bar{u}(c)\,u\big((d-1)/c\big)$ with $\bar{u}(c) = \left(\begin{smallmatrix} 1 & 0 \\ c & 1 \end{smallmatrix}\right)$, as one checks by multiplying out and using $ad - bc = 1$. Also $\bar{u}(c) = wu(-c)w^{-1}$ and $w^{-1} = w^3$. If c is not a unit, then a is a unit, because $ad \equiv 1 \pmod{\varpi}$. Then wk has lower row $(-a, -b)$, so wk is of the first kind, and $k = w^3(wk)$.

Reduction. Suppose that $s(gk) = s(g)s(k)$ for every generator g and every $k \in K$. Then induction on the length of a word gives $s(g_1 \cdots g_r k) = s(g_1) \cdots s(g_r)s(k)$. Taking $k = 1$ gives $s(g_1 \cdots g_r) = s(g_1) \cdots s(g_r)$, since $s(1) = (1,1)$ is the identity. So $s(k_1k_2) = s(k_1)s(k_2)$ for all k_1, k_2. In terms of κ, the condition $s(gk) = s(g)s(k)$ reads $\kappa(gk) = \kappa(g)\kappa(k)\beta(g,k)$.

The generator $u(x)$. The matrices $u(x)k$ and k have the same lower row, so $x(u(x)k) = x(k)$ and $\kappa(u(x)k) = \kappa(k)$. Also $\kappa(u(x)) = 1$, and $\beta(u(x), k) = (x(k)/1, x(k)/x(k))_n = 1$.

The generator w. Here $\kappa(w) = 1$, because the lower-left entry of w is a unit. The lower row of wk is $(-a, -b)$, and $\beta(w,k) = \big(-x(wk), x(wk)/x(k)\big)_n$, since $x(w) = -1$. We must show $\kappa(wk) = \kappa(k)\beta(w,k)$. There are four cases.

(i) c is a unit and a is 0 or a unit. Then $x(k) = c$, and $x(wk)$ is $-a$ or (if $a = 0$) $-b$, a unit since $bc = -1$. All arguments of the symbols are units, so both sides equal 1.

(ii) c is a unit and $0 < |a| < 1$. Then $x(k) = c$, $x(wk) = -a$, and b is a unit since $bc \equiv -1 \pmod{\varpi}$. Let $\alpha = v(a)$. By (H5), $\kappa(wk) = (-a,-b)_n = \omega(-b)^{-\alpha}$, and $\beta(w,k) = (a,-a/c)_n = \omega\big((-1)^{\alpha}(-c)^{\alpha}\big) = \omega(c)^{\alpha}$, since $a/(-a/c) = -c$. The two agree because $\omega(-b)^{-\alpha}\omega(c)^{-\alpha} = \omega(-bc)^{-\alpha} = 1$.

(iii) $c = 0$. Then a, d are units, $x(k) = d$, $x(wk) = -a$, and every argument is a unit. Both sides equal 1.

(iv) $0 < |c| < 1$. Then a, d are units with $ad \equiv 1 \pmod{\varpi}$, $x(k) = c$ and $x(wk) = -a$, so $\kappa(wk) = 1$. Let $\gamma = v(c)$. By (H5), $\kappa(k) = (c,d)_n = \omega(d)^{-\gamma}$ and $\beta(w,k) = (a,-a/c)_n = \omega(a)^{-\gamma}$. Their product is $\omega(ad)^{-\gamma} = 1$.

This proves the identity. The last assertion holds because $c = 0$ for upper triangular matrices. □

We fix this splitting. We write K also for $s(K)$ when no confusion can arise. The script `scripts/kubota_torus_check.py` in the repository also checks Lemma 3.3 numerically.

Example 3.4. Take $F = \mathbb{Q}_7$ and $n = 3$. By Lemma 3.2(a) and Example 3.1,

$$\tilde{h}(7)\tilde{h}(3) = (h(21),(3,7)_3), \qquad \tilde{h}(3)\tilde{h}(7) = (h(21),(7,3)_3).$$

The second coordinates are inverse to each other and are nontrivial cube roots of unity. So $\tilde{h}(7)$ and $\tilde{h}(3)$ do not commute, although $h(7)$ and $h(3)$ do.

3.3 Genuine representations and genuine functions

A genuine representation and a genuine test function transform by inverse characters of μ_n, so that their product is invariant and can be integrated over $\widetilde{G}$.

A representation π of $\widetilde{G}$ is *genuine* if $\pi(\zeta g) = \zeta\pi(g)$ for $\zeta \in \mu_n$. A function f on $\widetilde{G}$ is *genuine* if

$$f(\zeta g) = \zeta^{-1} f(g). \tag{3.3}$$

Let $C_c^{\infty}(\widetilde{G})_{\mathrm{gen}}$ be the space of genuine, locally constant, compactly supported functions. For a genuine admissible π and $f \in C_c^{\infty}(\widetilde{G})_{\mathrm{gen}}$ the operator $\pi(f) = \int_{\widetilde{G}} f(g)\pi(g)\,dg$ has finite rank. Its integrand is invariant under μ_n. The character Θ_π, where it is a function, satisfies $\Theta_\pi(\zeta g) = \zeta\Theta_\pi(g)$, and $\Theta_\pi f$ is invariant under μ_n. The same conventions apply to representations and functions on $\widetilde{A}$.

Example 3.5. Assume $\gcd(p, 2n) = 1$ and define the *genuine unit* e_K by $e_K(\zeta s(k)) = \zeta^{-1}$ for $k \in K$, and $e_K = 0$ off $\mu_n s(K)$. It is genuine, and it is invariant under conjugation by $s(K)$. For a genuine representation π and a vector v,

$$\pi(e_K)v = \int_{\mu_n}\int_K \zeta^{-1}\zeta\,\pi(s(k))v\,dk\,d\zeta = \int_K \pi(s(k))v\,dk.$$

So $\pi(e_K)$ is the projection onto the $s(K)$-fixed vectors, and $\operatorname{tr}\pi(e_K) = \dim \pi^{s(K)}$ when π is admissible.

3.4 Haar measures

Every identity below depends on Haar measures, so we fix them once and for all.

Give μ_n total mass one. On $A \cong F^\times$ use the Haar measure dt with $\mathrm{vol}(\mathscr{O}^\times) = 1$. On $N \cong F$ use du with $\mathrm{vol}(\mathscr{O}) = 1$. On K use dk with $\mathrm{vol}(K) = 1$. On $\widetilde{A} = \{(h(t), \zeta)\}$ use the product of dt and $d\zeta$. The measure on an open subgroup is the *restriction* of the measure on the group. We never replace it by a pushforward measure without saying so. For $\gcd(p, 2n) = 1$ we have $G = ANK$, hence $\widetilde{G} = \widetilde{A}Ns(K)$, and we normalize dg on $\widetilde{G}$ by

$$\int_{\widetilde{G}} \phi(g)\, dg = \int_{\widetilde{A}} \int_N \int_K \phi(a(u,1)\, s(k))\, da\, du\, dk. \tag{3.4}$$

The right side is a Haar measure on $\widetilde{G}$ by the standard argument for Iwasawa decompositions. With this normalization the set $\mu_n s(K)$ has volume one. On G we use the same formula without μ_n. The quotient $\widetilde{A}\backslash\widetilde{G}$ is identified with $A\backslash G$, and in both cases the quotient measure is $du\, dk$.

Example 3.6. Suppose $p \nmid d$, where d is a positive integer with $\mu_d \subset F$. The dth power map $\mathscr{O}^\times \to \mathscr{O}^\times$ has kernel μ_d and image $(\mathscr{O}^\times)^d$. Hence $(\mathscr{O}^\times)^d$ has index d in $\mathscr{O}^\times$, and its restricted volume is $1/d$. For instance, $(\mathbb{Z}_7^\times)^3$ has index 3 in $\mathbb{Z}_7^\times$ and volume $1/3$. This factor appears in Lemma 9.1.

3.5 Orbital integrals on $\mathrm{SL}(2, F)$

For the linear group, split orbital integrals are constant terms along N, and the Weyl integration formula expresses integrals over G through them. Section 6 generalizes both to the cover.

Let $a = h(t)$ with $t^2 \neq 1$. The *Weyl discriminant* and the *modulus character* are

$$D(a) = \det\big(1 - \mathrm{Ad}(a)\big)\big|_{\mathfrak{g}/\mathfrak{a}} = (1 - t^2)(1 - t^{-2}) = -(t - t^{-1})^2, \tag{3.5}$$

$$\delta_B(a) = |t|^2, \qquad |D(a)|^{1/2} = |t - t^{-1}|. \tag{3.6}$$

Here $\mathfrak{g}$ and $\mathfrak{a}$ are the Lie algebras of G and A. Both quantities pull back to $\widetilde{G}$ through ρ. For $f \in C_c^\infty(G)$ put $f^K(g) = \int_K f(k^{-1}gk)\, dk$. The *normalized split orbital integral* of f is

$$F_f^A(t) = |D(h(t))|^{1/2} \int_{A\backslash G} f(x^{-1}h(t)x)\, d\dot{x}, \qquad t^2 \neq 1. \tag{3.7}$$

The change of variables in the proof of Lemma 6.3 below, with $n = 1$, gives

$$F_f^A(t) = |t| \int_N f^K(h(t)u)\, du. \tag{3.8}$$

The right side is defined for all t, is locally constant and is compactly supported. It is the locally constant extension of F_f^A. It is invariant under $t \mapsto t^{-1}$.

The *Weyl integration formula* [Si79] states that for integrable ϕ on G,

$$\int_G \phi(g)\, dg = \frac{1}{2} \int_A |D(a)| \int_{A\backslash G} \phi(x^{-1}ax)\, d\dot{x}\, da + E(\phi), \tag{3.9}$$

where $E(\phi)$ is a finite sum of similar integrals over the elliptic maximal tori. It depends only on the restriction of ϕ to the set G_{ell} of regular elliptic elements.

Example 3.7. Let $f = 1_K$. Then $f^K = 1_K$. For a unit t, $h(t)u(x) \in K$ if and only if $x \in \mathscr{O}$. For a nonunit t, $h(t)u(x) \notin K$. By (3.8),

$$F^A_{1_K}(t) = 1_{\mathscr{O}^\times}(t).$$

3.6 Principal series of $\mathrm{SL}(2,F)$

The trace of a linear principal series is the integral of its inducing character against the split orbital integral. Theorem 9.4 compares this with the cover.

Let η be a smooth character of $A \cong F^\times$. The normalized principal series $\mathrm{Ind}_B^G(\eta\,\delta_B^{1/2})$ acts by right translation on locally constant $\phi : G \to \mathbb{C}$ with $\phi(bg) = \delta_B(b)^{1/2}\eta(b)\phi(g)$. Its character is

$$\Theta(h(t)) = \frac{\eta(t) + \eta(t)^{-1}}{|t - t^{-1}|} \quad (t^2 \neq 1), \qquad \Theta = 0 \text{ on } G_{\mathrm{ell}}, \tag{3.10}$$

and its trace is

$$\mathrm{tr}\,\mathrm{Ind}_B^G(\eta\,\delta_B^{1/2})(f) = \int_A \eta(t) F^A_f(t)\,dt. \tag{3.11}$$

Both formulas are the case $n = 1$ of Lemma 8.1, Corollary 8.2 and Theorem 8.3 below, so the paper proves them.

Example 3.8. Let η be unramified and $f = 1_K$. By Example 3.7 and (3.11), the trace is $\int_{\mathscr{O}^\times} \eta\,dt = 1$. This is the dimension of the space of K-fixed vectors.

4 Notation

The table collects the principal notation. Definitions are in Sections 3 and 5. For example, when $n = 6$ one has $d = 3$, $[\widetilde{A} : Z_A] = 9$ and $\dim \chi_\lambda = 3$.

Symbol	Meaning
$F, \mathscr{O}, \varpi, q, \lvert\cdot\rvert, v$	Local field, integers, uniformizer, residue field order, absolute value, valuation.
n, μ_n, $(\ ,\ )_n$	Degree of the cover, nth roots of unity, nth Hilbert symbol (Section 3.1).
$d = n/\gcd(n,2)$	Dimension of a genuine irreducible representation of $\widetilde{A}$.
G, $\widetilde{G} = \overline{SL(2,F)}$, ρ	$\mathrm{SL}(2,F)$, Kubota's n-fold cover, the projection.
$h(t), u(x), w$	Diagonal, unipotent and Weyl elements of G.
A, N, B, W	Diagonal torus, upper unipotent group, Borel subgroup, Weyl group.
$\widetilde{A}, \widetilde{B}$; $\tilde{h}(t), \tilde{w}$	Preimages of A and B. The lifts $(h(t),1)$ and $(w,1)$.
K, K_m, s, κ	$\mathrm{SL}(2,\mathscr{O})$, its principal congruence subgroups, the splitting (3.2).
Z_A, Z_A^0	Center of $\widetilde{A}$, equal to $\rho^{-1}(h((F^\times)^d))$. The set $\{(h(u),1) : u \in (F^\times)^d\}$.
C, $\widetilde{C}$	$h(\varpi^{d\mathbb{Z}}\mathscr{O}^\times)$ and its preimage, a maximal abelian subgroup of $\widetilde{A}$.
λ, $\tilde{\lambda}$, χ_λ	Genuine character of Z_A, an extension to $\widetilde{C}$, the induced representation of $\widetilde{A}$.
λ^w, $\chi^{\tilde{w}}$	Weyl conjugates $z \mapsto \lambda(\tilde{w}^{-1} z \tilde{w})$ and $a \mapsto \chi(\tilde{w}^{-1} a \tilde{w})$.
D, δ_B	Weyl discriminant and modulus character.
f^K, F_f^A	K-average and linear split orbital integral.
Ψ_f, Φ_f, $\mathscr{O}_f^{Z_A}$	Constant term on $\widetilde{A}$ and normalized split orbital integrals on Z_A and $\widetilde{A}$ (Section 6).
e_K	Genuine unit (Example 3.5).
$\mathrm{I}(\chi)$, Θ_Π	Normalized genuine principal series and its character.
ν, η_λ	Power map $h(t) \mapsto (h(t^d),1)$ and the character $t \mapsto \lambda(\nu(h(t)))$ of A.

5 The covering torus

The commutator of two lifts of diagonal elements is a power of the Hilbert symbol. Its radical is the center Z_A, and a Stone–von Neumann argument parametrizes the genuine irreducible representations of $\widetilde{A}$ by genuine characters of Z_A.

5.1 The commutator, the center and Weyl conjugation

We compute the commutator pairing on $\widetilde{A}$, its radical, and the action of $\tilde{w}$.

Lemma 5.1. *For $a, b \in F^\times$,*

$$\tilde{h}(a)\tilde{h}(b)\tilde{h}(a)^{-1}\tilde{h}(b)^{-1} = (a,b)_n^{-2}. \tag{5.1}$$

Proof. By Lemma 3.2(a), $\tilde{h}(a)\tilde{h}(b) = (h(ab), (b,a)_n)$ and $\tilde{h}(b)\tilde{h}(a) = (h(ab), (a,b)_n)$. Hence $\tilde{h}(a)\tilde{h}(b) = (b,a)_n(a,b)_n^{-1}\tilde{h}(b)\tilde{h}(a) = (a,b)_n^{-2}\tilde{h}(b)\tilde{h}(a)$ by (H2). □

Define

$$Z_A = \rho^{-1}\big(h((F^\times)^d)\big), \qquad d = \frac{n}{\gcd(n,2)}. \tag{5.2}$$

This is a set of the form $\mu_n \cdot Z_A^0$ with $Z_A^0 = \{(h(u),1) : u \in (F^\times)^d\}$. It is not the set of dth powers of elements of $\widetilde{A}$.

Lemma 5.2. *The group Z_A is the center of $\widetilde{A}$. If $\gcd(p,n)=1$, then*

$$[\widetilde{A} : Z_A] = d^2, \tag{5.3}$$

and the commutator induces a nondegenerate alternating pairing on the finite group $\widetilde{A}/Z_A$.

Proof. Since μ_n is central, $\tilde{h}(a)$ is central in $\widetilde{A}$ exactly when $(a,b)_n^2 = 1$ for all b, by Lemma 5.1. By (H1) this says $(a^2,b)_n = 1$ for all b. By (H4) it says $a^2 \in (F^\times)^n$. We show that this holds exactly when $a \in (F^\times)^d$.

If n is odd, then $d = n$. Choose integers k,m with $2k+nm = 1$. Then $a = (a^2)^k (a^n)^m$, so $a^2 \in (F^\times)^n$ implies $a \in (F^\times)^n$. The converse is clear.

If n is even, then $n = 2d$. If $a^2 = c^{2d}$, then $(a/c^d)^2 = 1$, so $a = \pm c^d$. A primitive nth root of unity ζ lies in F, and $\zeta^d = -1$. So $-1 \in (F^\times)^d$ and $a \in (F^\times)^d$. Conversely, $a = c^d$ gives $a^2 = c^n$.

Now let $p \nmid n$. Since $\mu_n \subset F$ and $p \nmid n$, the group μ_n injects into the residue field, so $n \mid q-1$ and $d \mid q-1$. We have $F^\times \cong \varpi^{\mathbb{Z}} \times \mu_{q-1} \times (1+\varpi\mathscr{O})$. The dth power map is bijective on the pro-p group $1+\varpi\mathscr{O}$. Hence $F^\times/(F^\times)^d \cong \mathbb{Z}/d \times \mathbb{Z}/d$, which gives (5.3). The pairing $(a,b) \mapsto (a,b)_n^{-2}$ is alternating because $(a,a)_n^2 = (a,-1)_n^2 = 1$. It is nondegenerate on $\widetilde{A}/Z_A$ by the first part of the proof. □

Remark 5.3. The centralizer of all of $\widetilde{A}$ is Z_A. The centralizer in $\widetilde{G}$ of a single regular element $a \in \widetilde{A}$ can be larger than Z_A and smaller than $\widetilde{A}$. So one may not integrate over $\widetilde{A}\backslash\widetilde{G}$ in an orbital integral unless a commutes with all of $\widetilde{A}$, that is, unless $a \in Z_A$.

Lemma 5.4. *For $t \in F^\times$:*

(a) $\tilde{w}^{-1} = (w^{-1}, 1)$.

(b) $\tilde{w}^{-1}\tilde{h}(t)\tilde{w} = \tilde{h}(t^{-1})$.

(c) $\tilde{h}(t)^{-1} = (t,-1)_n \tilde{h}(t^{-1})$.

(d) $\tilde{w}^2$ lies in the preimage of $-I$, so it is central in $\widetilde{G}$.

Proof. (a) We have $x(w) = -1$, $x(w^{-1}) = 1$ and $x(I) = 1$, so $\beta(w,w^{-1}) = (-1,1)_n = 1$.

(b) The matrix $w^{-1}h(t)$ has lower row $(t,0)$. So $x(w^{-1}h(t)) = t$, while $x(h(t)) = t^{-1}$. Thus $\beta(w^{-1},h(t)) = (t,t^2)_n = (t,t)_n^2 = (t,-1)_n^2 = 1$. Next, $w^{-1}h(t)w = h(t^{-1})$ and $x(h(t^{-1})) = t$. Thus

$$\beta(w^{-1}h(t),w) = \left(\frac{t}{t},\ \frac{t}{-1}\right)_n = (1,-t)_n = 1.$$

Multiplying out, $(w^{-1},1)(h(t),1)(w,1) = (h(t^{-1}),1)$.

(c) By Lemma 3.2(a), $\tilde{h}(t)\tilde{h}(t^{-1}) = (I,(t^{-1},t)_n)$ and $(t^{-1},t)_n = (t,t)_n^{-1} = (t,-1)_n^{-1} = (t,-1)_n$. Hence $\tilde{h}(t)^{-1} = (t,-1)_n \tilde{h}(t^{-1})$.

(d) $w^2 = -I$, and Lemma 3.2(d) applies. □

By (b), Weyl conjugation and inversion agree on the matrix part of $\tilde{h}(t)$. They differ on μ_n: $\tilde{w}^{-1}\zeta\tilde{h}(t)\tilde{w} = \zeta\tilde{h}(t^{-1})$, whereas $(\zeta\tilde{h}(t))^{-1} = \zeta^{-1}(t,-1)_n\tilde{h}(t^{-1})$.

Example 5.5. Take $F = \mathbb{Q}_5$, $n = 4$ and $t = 5$. By Example 3.1, $(5,-1)_4 = -1$. So $\tilde{h}(5)^{-1} = -\tilde{h}(1/5)$, while $\tilde{w}^{-1}\tilde{h}(5)\tilde{w} = \tilde{h}(1/5)$. Even on the lifts $\tilde{h}(t)$, inversion and Weyl conjugation differ.

5.2 The Stone–von Neumann parametrization

Because the commutator pairing on $\widetilde{A}/Z_A$ is nondegenerate, a maximal abelian subgroup has index d, and every genuine irreducible representation is induced from one of its characters. This is the rank-one analogue of the torus theory of [Mc12]. Since that work assumes $\mu_{2n} \subset F$, we give the argument in full. Assume $\gcd(p,n) = 1$ and put

$$C = h(\varpi^{d\mathbb{Z}}\mathscr{O}^\times), \qquad \widetilde{C} = \rho^{-1}(C).$$

Two elements $h(\varpi^{dk}u)$ and $h(\varpi^{dl}u')$ of C have commutator $(\varpi^{dk}u, \varpi^{dl}u')_n^{-2}$. By (H1) and (H5) this equals $(\varpi,u')_n^{-2dk}(u,\varpi)_n^{-2dl}(\varpi,\varpi)_n^{-2d^2kl}$. Each factor is an nth root of unity raised to a multiple of $2d$, which is a multiple of n. So $\widetilde{C}$ is abelian. It contains Z_A, and $[\widetilde{A} : \widetilde{C}] = [F^\times : \varpi^{d\mathbb{Z}}\mathscr{O}^\times] = d$. Since $d^2 = [\widetilde{A} : Z_A]$ and the pairing on $\widetilde{A}/Z_A$ is nondegenerate, $\widetilde{C}$ is a maximal abelian subgroup.

Theorem 5.6 (Genuine representations of $\widetilde{A}$)**.** *Assume* $\gcd(p,n) = 1$. *For each smooth genuine character* $\lambda : Z_A \to \mathbb{C}^\times$ *choose an extension* $\tilde{\lambda}$ *to* $\widetilde{C}$. *Then*

$$\chi_\lambda = \operatorname{Ind}_{\widetilde{C}}^{\widetilde{A}} \tilde{\lambda} \tag{5.4}$$

is irreducible of dimension d. *Its isomorphism class does not depend on the extension. The map* $\lambda \mapsto \chi_\lambda$ *is a bijection from genuine smooth characters of* Z_A *to genuine smooth irreducible representations of* $\widetilde{A}$. *The trace is*

$$\operatorname{tr}\chi_\lambda(a) = \begin{cases} d\,\lambda(a), & a \in Z_A, \\ 0, & a \in \widetilde{A} \setminus Z_A. \end{cases} \tag{5.5}$$

Proof. An extension exists because $\widetilde{C}/Z_A$ is a finite abelian group and $\mathbb{C}^\times$ is divisible. Every extension is smooth, because Z_A is open in $\widetilde{C}$ and λ is smooth. For $b \in \widetilde{A}$ the conjugate extension is $\tilde{\lambda}^b(c) = \tilde{\lambda}(b^{-1}cb) = \langle b,c\rangle\,\tilde{\lambda}(c)$, where $\langle b,c\rangle = b^{-1}cbc^{-1} \in \mu_n$ is a commutator. The map $b \mapsto \langle b,\cdot\rangle$ embeds $\widetilde{A}/\widetilde{C}$ into the character group of $\widetilde{C}/Z_A$, by maximality of $\widetilde{C}$. Both groups have order d. So the conjugates of $\tilde{\lambda}$ are exactly the d extensions of λ, and the stabilizer of $\tilde{\lambda}$ is $\widetilde{C}$. Mackey's criterion shows that χ_λ is irreducible of dimension $[\widetilde{A} : \widetilde{C}] = d$, and that its class does not depend on the extension.

Conversely, let σ be a genuine smooth irreducible representation of $\widetilde{A}$. By Schur's lemma Z_A acts by a genuine character λ. Since $\widetilde{C}/Z_A$ is finite and abelian, the restriction of σ to $\widetilde{C}$ contains a character $\tilde{\lambda}$ extending λ. Frobenius reciprocity gives a nonzero map $\chi_\lambda \to \sigma$, which is an isomorphism because both are irreducible.

For $a \in Z_A$, Schur's lemma gives $\chi_\lambda(a) = \lambda(a)$ times the identity, so the trace is $d\lambda(a)$. If $a \notin Z_A$, choose $b \in \widetilde{A}$ with $b^{-1}ab = \zeta a$ and $\zeta \neq 1$. Then $\operatorname{tr}\chi_\lambda(a) = \operatorname{tr}\chi_\lambda(b^{-1}ab) = \zeta\operatorname{tr}\chi_\lambda(a)$, so the trace vanishes. □

Example 5.7 (The double and triple covers)**.** For $n = 2$ we have $d = 1$. The group $\widetilde{A}$ is abelian, $Z_A = \widetilde{A}$, and the genuine irreducible representations of $\widetilde{A}$ are characters. Still, $t \mapsto \tilde{h}(t)$ is not a homomorphism, since $\beta(h(a),h(b)) = (b,a)_2$ need not be 1.

For $n = 3$ we have $d = 3$ and $[\widetilde{A} : Z_A] = 9$. Each genuine irreducible representation of $\widetilde{A}$ is three-dimensional and is induced from a character of a subgroup of index three. Its trace is supported on Z_A and equals three times the central character there. By Example 3.4, with $F = \mathbb{Q}_7$, the element $\tilde{h}(7)$ is not in Z_A, and its trace is zero.

Example 5.8 (Degrees four and six)**.** For $n = 4$ we have $d = 2$ and $[\widetilde{A} : Z_A] = 4$. The genuine irreducible representations of $\widetilde{A}$ are two-dimensional. The map $t \mapsto (h(t^2), 1)$ is a homomorphism, because $\beta(h(a^2), h(b^2)) = (b,a)_4^4 = 1$. So Theorem 9.4 applies in degree four.

For $n = 6$ we have $d = 3$ and $[\widetilde{A} : Z_A] = 9$. Now $\beta(h(a^3), h(b^3)) = (b,a)_6^9 = (b,a)_6^3$, which need not be 1. Theorem 5.6 still holds, but the map $t \mapsto (h(t^3), 1)$ is not a homomorphism. This is why Theorem 9.4 excludes $n = 6$.

6 Split orbital integrals and their image

Only elements of the center Z_A have nonzero normalized orbital integrals. On Z_A these integrals fill out exactly the Weyl-invariant genuine test functions. That image theorem is what makes the comparison with $\mathrm{SL}(2,F)$ in Section 9 possible.

6.1 The correct quotient

Because $\widetilde{A}$ is not abelian, the group divided out in an orbital integral must centralize the element (Remark 5.3).

Since Z_A centralizes $\widetilde{A}$, the integral

$$\mathscr{O}_f^{Z_A}(a) = |D(a)|^{1/2} \int_{Z_A \backslash \widetilde{G}} f(x^{-1} a x)\, d\dot{x}, \qquad a \in \widetilde{A}_{\mathrm{reg}}, \tag{6.1}$$

is well defined for $f \in C_c^\infty(\widetilde{G})_{\mathrm{gen}}$. Here $\widetilde{A}_{\mathrm{reg}} = \rho^{-1}(A_{\mathrm{reg}})$ and $A_{\mathrm{reg}} = \{h(t) : t^2 \neq 1\}$. For $z \in (Z_A)_{\mathrm{reg}} = Z_A \cap \widetilde{A}_{\mathrm{reg}}$ the centralizer of z in $\widetilde{G}$ is $\widetilde{A}$, so we also set

$$\Phi_f(z) = |D(z)|^{1/2} \int_{\widetilde{A} \backslash \widetilde{G}} f(x^{-1} z x)\, d\dot{x}. \tag{6.2}$$

The measure on Z_A is the restriction of the measure on $\widetilde{A}$, so $Z_A \backslash \widetilde{A}$ has volume $[\widetilde{A} : Z_A]$. Hence

$$\mathscr{O}_f^{Z_A}(z) = d^2\, \Phi_f(z) \qquad (z \in (Z_A)_{\mathrm{reg}}) \tag{6.3}$$

when $\gcd(p,n) = 1$. The index d^2 should not be confused with d, the dimension of χ_λ.

Three quotients. Three subgroups of $\widetilde{A}$ centralize an element $z \in (Z_A)_{\mathrm{reg}}$: the center Z_A, the maximal abelian subgroup $\widetilde{C}$, and $\widetilde{A}$ itself. Since $Z_A \subset \widetilde{C} \subset \widetilde{A}$ with $[\widetilde{C} : Z_A] = [\widetilde{A} : \widetilde{C}] = d$, the three orbital integrals of f at z differ only by these indices:

quotient	index over $\widetilde{A}\backslash\widetilde{G}$	normalized integral at z
$Z_A\backslash\widetilde{G}$	d^2	$\mathscr{O}_f^{Z_A}(z) = d^2\,\Phi_f(z)$
$\widetilde{C}\backslash\widetilde{G}$	d	$\mathscr{O}_f^{\widetilde{C}}(z) = d\,\Phi_f(z)$
$\widetilde{A}\backslash\widetilde{G}$	1	$\Phi_f(z)$

(6.4)

Here $\mathscr{O}_f^{\widetilde{C}}$ is defined like (6.1) with $\widetilde{C}$ in place of Z_A. For $a \in \widetilde{A}_{\mathrm{reg}} \setminus Z_A$ the Z_A-quotient integral is still defined, and the $\widetilde{C}$-quotient integral is defined when $a \in \widetilde{C}$. Both vanish there, by Lemma 6.1 and the same proof, since $\widetilde{C}$ is normal in $\widetilde{A}$.

Lemma 6.1 (Vanishing). *For $f \in C_c^\infty(\widetilde{G})_{\mathrm{gen}}$ and $a \in \widetilde{A}_{\mathrm{reg}} \setminus Z_A$, $\mathscr{O}_f^{Z_A}(a) = 0$.*

Proof. By Lemma 5.2 there is $b \in \widetilde{A}$ with $b^{-1}ab = \zeta a$ and $\zeta \neq 1$. Left translation $x \mapsto bx$ is well defined on $Z_A\backslash\widetilde{G}$, because b centralizes Z_A. It preserves the quotient measure, because $\widetilde{A}$ and $\widetilde{G}$ are unimodular. After the substitution the integrand is $f(x^{-1}b^{-1}abx) = f(\zeta x^{-1}ax) = \zeta^{-1}f(x^{-1}ax)$. Hence $\mathscr{O}_f^{Z_A}(a) = \zeta^{-1}\mathscr{O}_f^{Z_A}(a)$, and the integral is zero. □

Example 6.2. Take $F = \mathbb{Q}_7$, $n = 3$ and $a = \tilde{h}(7)$, which is regular. By Example 3.4, $b = \tilde{h}(3)$ satisfies $b^{-1}ab = \zeta a$ with ζ a nontrivial cube root of unity. So $\mathscr{O}_f^{Z_A}(\tilde{h}(7)) = 0$ for every genuine f.

6.2 The constant term and the image theorem

Writing Φ_f as a constant term along N extends it across the singular set and leads to the image theorem. Assume $\gcd(p,2n) = 1$ and use the splitting s of K. For $f \in C_c^\infty(\widetilde{G})_{\mathrm{gen}}$ put $f^K(g) = \int_K f(s(k)^{-1}gs(k))\,dk$, and for $a \in \widetilde{A}$ put

$$\Psi_f(a) = \delta_B(a)^{1/2} \int_N f^K(a\,(u,1))\,du. \tag{6.5}$$

Lemma 6.3. *Assume* $\gcd(p,2n) = 1$. *Then Ψ_f is locally constant and compactly supported on $\widetilde{A}$, and $\Psi_f(\zeta a) = \zeta^{-1}\Psi_f(a)$. For $z \in (Z_A)_{\mathrm{reg}}$,*

$$\Phi_f(z) = \Psi_f(z) = \delta_B(z)^{1/2} \int_N f^K(z\,(u,1))\,du. \tag{6.6}$$

Proof. The first assertions follow because f^K is genuine, locally constant and compactly supported. Let $z \in (Z_A)_{\mathrm{reg}}$ with $\rho(z) = h(t)$. By (3.4) the quotient measure on $\widetilde{A}\backslash\widetilde{G}$ is $du\,dk$, so

$$\Phi_f(z) = |D(z)|^{1/2} \int_N f^K\big((u,1)^{-1}z(u,1)\big)du.$$

Write $u = u(x)$. By Lemma 3.2(c), $z^{-1}(u(x),1)z$ is the canonical lift of $h(t)^{-1}u(x)h(t) = u(t^{-2}x)$. Hence $(u(x),1)^{-1}z(u(x),1) = z(u((1-t^{-2})x),1)$. Substitute $y = (1-t^{-2})x$, so that $dx = |1-t^{-2}|^{-1}dy$. Since $|D(z)|^{1/2} = |t|\,|1-t^{-2}|$ and $\delta_B(z)^{1/2} = |t|$, we obtain (6.6). □

We use $\Psi_f|_{Z_A}$ as the locally constant extension of Φ_f to all of Z_A, and we keep the name Φ_f. On $(Z_A)_{\mathrm{reg}}$, Φ_f is invariant under $z \mapsto \tilde{w}^{-1}z\tilde{w}$. Indeed $\tilde{w}$ normalizes $\widetilde{A}$, so $x \mapsto \tilde{w}x$ preserves the measure on $\widetilde{A}\backslash\widetilde{G}$, and $|D|$ is invariant. By local constancy the extension is also invariant. Let $C_c^\infty(Z_A)_{\mathrm{gen}}^W$ be the space of locally constant, compactly supported functions ϕ on Z_A with $\phi(\zeta z) = \zeta^{-1}\phi(z)$ and $\phi(\tilde{w}^{-1}z\tilde{w}) = \phi(z)$. By Lemma 5.4(b) the second condition reads $\phi(\zeta\tilde{h}(u^{-1})) = \phi(\zeta\tilde{h}(u))$.

Theorem 6.4 (Image of the split orbital integral). *Assume* $\gcd(p,2n) = 1$. *The map*

$$C_c^\infty(\widetilde{G})_{\mathrm{gen}} \longrightarrow C_c^\infty(Z_A)_{\mathrm{gen}}^W, \qquad f \longmapsto \Phi_f, \tag{6.7}$$

is well defined and surjective.

Proof. Lemma 6.3 and the remarks after it show that the map lands in $C_c^\infty(Z_A)^W_{\mathrm{gen}}$. We prove surjectivity in two steps.

Step 1: functions supported on the regular set. Consider

$$\mu : (\widetilde{A}\backslash\widetilde{G}) \times (Z_A)_{\mathrm{reg}} \to \widetilde{G}, \qquad (\widetilde{A}x, z) \mapsto x^{-1}zx.$$

It is well defined because z is central in $\widetilde{A}$. Its composition with ρ is the map $(Ax, h(t)) \mapsto x^{-1}h(t)x$. At a regular point the differential of that map is bijective, since $\mathrm{Ad}(h(t)) - 1$ is invertible on $\mathfrak{g}/\mathfrak{a}$. By the p-adic inverse function theorem it is a local homeomorphism onto an open set. Since ρ is a local homeomorphism, so is μ.

We describe the fibers of μ. Suppose $x^{-1}zx = y^{-1}z'y$ and put $c = yx^{-1}$. Then c conjugates z to z'. Both are regular, so $\rho(c)$ normalizes A and $c \in \widetilde{A} \cup \tilde{w}\widetilde{A}$. If $c \in \widetilde{A}$, then $z' = z$ and $\widetilde{A}y = \widetilde{A}x$. If $c = \tilde{w}b$ with $b \in \widetilde{A}$, then $z' = \tilde{w}z\tilde{w}^{-1}$ and $\widetilde{A}y = \widetilde{A}\tilde{w}x$. So each fiber is $\{(\widetilde{A}x, z), (\widetilde{A}\tilde{w}x, \tilde{w}z\tilde{w}^{-1})\}$.

Let α be the characteristic function of $\widetilde{A}s(K)$, viewed on $\widetilde{A}\backslash\widetilde{G}$. By (3.4), $\int \alpha = 1$. Let $\phi \in C_c^\infty(Z_A)^W_{\mathrm{gen}}$ vanish near the singular set. Define f on the image of μ by

$$f(x^{-1}zx) = \phi(z)\big(\alpha(\widetilde{A}x) + \alpha(\widetilde{A}\tilde{w}x)\big),$$

and $f = 0$ elsewhere. The value at the second point of the fiber is $\phi(\tilde{w}z\tilde{w}^{-1})(\alpha(\widetilde{A}\tilde{w}x) + \alpha(\widetilde{A}\tilde{w}^2x))$. It is the same, because ϕ is Weyl invariant and $\tilde{w}^2 \in \widetilde{A}$. So f is well defined. It is genuine because ϕ is.

We check that $f \in C_c^\infty(\widetilde{G})$. Let $S = \operatorname{supp}\alpha \cup \tilde{w}^{-1}\operatorname{supp}\alpha$, a compact open subset of $\widetilde{A}\backslash\widetilde{G}$, and let $\Sigma = \operatorname{supp}\phi$, a compact open subset of $(Z_A)_{\mathrm{reg}}$. The function $f \circ \mu$ is locally constant and vanishes off $S \times \Sigma$. Since μ is a local homeomorphism, f is locally constant on the open set $U = \mu((\widetilde{A}\backslash\widetilde{G}) \times (Z_A)_{\mathrm{reg}})$. The support of f lies in $\mu(S \times \Sigma)$, which is compact because μ is continuous. A compact subset of the open set U has a neighborhood on which f is defined by the formula, and f vanishes on the open complement of $\mu(S \times \Sigma)$. These two open sets cover $\widetilde{G}$, and f is locally constant on each. So f is locally constant on $\widetilde{G}$, and it has compact support. Finally, the left translation $x \mapsto \tilde{w}x$ preserves the measure on $\widetilde{A}\backslash\widetilde{G}$, so

$$\Phi_f(z) = |D(z)|^{1/2}\phi(z)\int \big(\alpha(\widetilde{A}x) + \alpha(\widetilde{A}\tilde{w}x)\big)d\dot{x} = 2|D(z)|^{1/2}\phi(z).$$

Applying this to $\phi/(2|D|^{1/2})$, which lies in the same space, gives a preimage of ϕ.

Step 2: the singular set. The singular elements of Z_A lie over $\pm I$. For $\varepsilon = \pm 1$ let $z_\varepsilon = (\varepsilon I, 1)$, which is central by Lemma 3.2(d), and lies in Z_A because $-1 \in (F^\times)^d$ (proof of Lemma 5.2). For $m \geq 1$ let $f_{\varepsilon,m}(\zeta z_\varepsilon s(k)) = \zeta^{-1}$ for $k \in K_m$, and $f_{\varepsilon,m} = 0$ off $\mu_n z_\varepsilon s(K_m)$. This function is genuine, locally constant and compactly supported. It is invariant under conjugation by $s(K)$, because z_ε is central and K_m is normal in K. So $f_{\varepsilon,m}^K = f_{\varepsilon,m}$.

We compute $\Phi_{f_{\varepsilon,m}}$ on all of Z_A. Let $z \in Z_A$ and $x \in F$. If $z(u(x), 1) \in \mu_n z_\varepsilon s(K_m)$, then $\rho(z)u(x) \in \varepsilon K_m$, which forces $\rho(z) \in \varepsilon h(1 + \varpi^m\mathscr{O})$ and $x \in \varpi^m\mathscr{O}$. So $\Phi_{f_{\varepsilon,m}}$ vanishes off the set $U_{\varepsilon,m} = \mu_n z_\varepsilon\{(h(t), 1) : t \in 1 + \varpi^m\mathscr{O}\}$. Every element of $U_{\varepsilon,m}$ has the form $z = \zeta z_\varepsilon(h(t), 1)$. By Lemma 3.2(b) and Lemma 3.3, $z(u(x), 1) = \zeta z_\varepsilon s(h(t)u(x))$, and this lies in $\mu_n z_\varepsilon s(K_m)$ exactly when $x \in \varpi^m\mathscr{O}$. By (6.6),

$$\Phi_{f_{\varepsilon,m}} = q^{-m}\mathbf{1}^{\mathrm{gen}}_{U_{\varepsilon,m}} \quad \text{on all of } Z_A,$$

where $\mathbf{1}_U^{\mathrm{gen}}(\zeta z') = \zeta^{-1}$ for $\zeta z' \in U$ with $z' \in z_\varepsilon\{(h(t),1)\}$, and $\mathbf{1}_U^{\mathrm{gen}} = 0$ off U. In the tame case $1+\varpi^m\mathscr{O} \subset (F^\times)^d$, so $U_{\varepsilon,m}$ is an open neighborhood of z_ε in Z_A. It is Weyl stable, by Lemma 5.4(b). Since p is odd, $-1 \notin 1+\varpi\mathscr{O}$, so $U_{1,m}$ and $U_{-1,m}$ are disjoint for every $m \geq 1$.

Now let $\phi \in C_c^\infty(Z_A)^W_{\mathrm{gen}}$ be arbitrary. Choose m so large that $\phi(\zeta z_\varepsilon(h(t),1)) = \zeta^{-1}\phi(z_\varepsilon)$ for $t \in 1+\varpi^m\mathscr{O}$ and $\varepsilon = \pm 1$. Then

$$\phi - \sum_{\varepsilon=\pm 1} q^m \phi(z_\varepsilon)\,\Phi_{f_{\varepsilon,m}}$$

vanishes near the singular set and lies in $C_c^\infty(Z_A)^W_{\mathrm{gen}}$. Step 1 gives a preimage of it, and so ϕ has a preimage. □

Remark 6.5 (Comparison with the 1999 and 2001 normalizations). The 1999 manuscript writes the transform as $F_f^{\overline{A}}$ on the preimage of $A^{n_0} = h((F^\times)^d)$, which is Z_A, with the quotient $\overline{A}\backslash\overline{G}$. With the measures of Section 3.4 it is our Φ_f. If an orbital integral is formed instead with the quotient by Z_A or by the maximal abelian subgroup $\widetilde{C}$, then (6.4) gives the conversion: multiply by d^2 or by d. In each normalization the image is the same space $C_c^\infty(Z_A)^W_{\mathrm{gen}}$. The image theorem of [Jo01] uses yet another notation, and its quotient and measures must be matched by the same table before the two statements are identified.

Example 6.6. Assume $\gcd(p,2n) = 1$ and let e_K be the genuine unit of Example 3.5. Every element of Z_A is $\zeta(h(u),1)$ with $u \in (F^\times)^d$. For $u \in \mathscr{O}^\times$ and $x \in \mathscr{O}$, $(h(u),1)(u(x),1) = s(h(u)u(x))$ by Lemma 3.2(b). For $u \notin \mathscr{O}^\times$ the matrix $h(u)u(x)$ is not in K. Since $e_K^K = e_K$, (6.6) gives

$$\Phi_{e_K}(\zeta(h(u),1)) = \zeta^{-1}1_{\mathscr{O}^\times}(u) \qquad (u \in (F^\times)^d).$$

Note that $(F^\times)^d \cap \mathscr{O}^\times = (\mathscr{O}^\times)^d$.

Remark 6.7 (Satake and the image theorem). Formula (6.6) is a constant term of the conjugation average of a test function. It should not be confused with the spherical Satake isomorphism of convolution algebras [Mc12]. Surjectivity of (6.7) on all test functions does not by itself give a statement about spherical functions.

7 Principal series and known unitary representations

Genuine principal series are induced from the d-dimensional representations χ_λ of $\widetilde{A}$. We compute their Jacquet modules and prove unitarity, an irreducibility criterion and the structure of the spherical vectors. We then survey the literature on reducibility, complementary series and supercuspidal representations. Statements proved elsewhere are in remarks numbered "Remark 2026.n" with their status. The examples treat the threefold cover.

7.1 Genuine principal series

We define the representations and compute their Jacquet modules directly, so that nothing depends on the general theory of [Mc12], whose hypotheses are stronger than ours.

Let χ be a genuine smooth finite-dimensional representation of $\widetilde{A}$ on V_χ, extended to $\widetilde{B} = \widetilde{A}N$ trivially on the image of N. The *normalized principal series* $\mathrm{I}(\chi) = \mathrm{Ind}_{\widetilde{B}}^{\widetilde{G}}(\chi \otimes \delta_B^{1/2})$ acts by right translation on locally constant $\phi : \widetilde{G} \to V_\chi$ with

$$\phi(bg) = \delta_B(b)^{1/2}\chi(b)\phi(g), \qquad b \in \widetilde{B},\ g \in \widetilde{G}. \tag{7.1}$$

It is genuine and admissible. Put

$$\chi^{\tilde{w}}(a) = \chi(\tilde{w}^{-1}a\tilde{w}). \tag{7.2}$$

By Lemma 5.4, $\chi^{\tilde{w}}$ is not $a \mapsto \chi(a^{-1})$ in general. For a smooth representation V of $\widetilde{B}$ let V_N be its space of coinvariants under the image of N, and let $r_N(V)$ be V_N with the action of $\widetilde{A}$ twisted by $\delta_B^{-1/2}$. This is the normalized Jacquet module. We use two general facts about ℓ-groups [BZ76, BZ77]. First, $V \mapsto V_N$ is exact, because N is a union of compact open subgroups. Second, $C_c^\infty(N)_N \cong \mathbb{C}$ through $\phi \mapsto \int_N \phi$. We also use Frobenius reciprocity $\mathrm{Hom}_{\widetilde{G}}(\pi, \mathrm{I}(\chi)) \cong \mathrm{Hom}_{\widetilde{A}}(r_N(\pi), \chi)$, which holds for parabolic induction in any ℓ-group of this form [BZ77].

Lemma 7.1 (Jacquet module)**.** *Let χ be a genuine smooth finite-dimensional representation of $\widetilde{A}$. Then $r_N(\mathrm{I}(\chi))$ has a subrepresentation isomorphic to $\chi^{\tilde{w}}$ with quotient isomorphic to χ.*

Proof. Let $V = \mathrm{I}(\chi)$ and $V_0 = \{\phi \in V : \phi(1) = 0\}$. Since $G = B \sqcup BwN$, we have $\widetilde{G} = \widetilde{B} \sqcup \widetilde{B}\tilde{w}N$, and $\widetilde{B}\backslash\widetilde{G}$ is compact. A function in V_0 vanishes on $\widetilde{B}$ and is locally constant, so it vanishes near $\widetilde{B}$. Hence $\phi \mapsto (u \mapsto \phi(\tilde{w}(u,1)))$ is an isomorphism from V_0 onto $C_c^\infty(N, V_\chi)$, and $\phi \mapsto \phi(1)$ identifies V/V_0 with V_χ. In both cases the image of N acts by right translation.

On V/V_0, the element $a \in \widetilde{A}$ acts by $\delta_B(a)^{1/2}\chi(a)$, so $r_N(V/V_0) \cong \chi$. On V_0, write $a = \zeta\tilde{h}(t)$. Then $\tilde{w}(u,1)a = (\tilde{w}a\tilde{w}^{-1})\,\tilde{w}(a^{-1}(u,1)a)$, and conjugation by a preserves the canonical lift of N (Lemma 3.2(c)). So a sends $F(u) = \phi(\tilde{w}(u,1))$ to $\delta_B(\tilde{w}a\tilde{w}^{-1})^{1/2}\chi(\tilde{w}a\tilde{w}^{-1})F(a^{-1}ua)$. Integrating over N, and using $\int_N F(a^{-1}ua)\,du = \delta_B(a)\int_N F(u)\,du$ and $\delta_B(\tilde{w}a\tilde{w}^{-1}) = \delta_B(a)^{-1}$, we find that a acts on $(V_0)_N \cong V_\chi$ by $\delta_B(a)^{1/2}\chi(\tilde{w}a\tilde{w}^{-1})$. After the twist by $\delta_B^{-1/2}$ this is $\chi(\tilde{w}a\tilde{w}^{-1})$. Since $\tilde{w}^2$ is central (Lemma 5.4(d)), $\chi(\tilde{w}a\tilde{w}^{-1}) = \chi(\tilde{w}^{-1}(\tilde{w}^2a\tilde{w}^{-2})\tilde{w}) = \chi^{\tilde{w}}(a)$. Exactness of r_N gives the statement. □

For $\chi = \chi_\lambda$ we have $\chi^{\tilde{w}} \cong \chi_{\lambda^w}$ with $\lambda^w(z) = \lambda(\tilde{w}^{-1}z\tilde{w})$, by Theorem 5.6. Call χ_λ *Weyl regular* if $\lambda^w \neq \lambda$.

Proposition 7.2. *Let $\chi = \chi_\lambda$ with λ unitary.*

(a) $\mathrm{I}(\chi)$ *is unitary.*

(b) If χ is Weyl regular, then $\mathrm{I}(\chi)$ *is irreducible.*

Proof. (a) Let $\langle\ ,\ \rangle$ be a $\widetilde{A}$-invariant inner product on V_χ. For ϕ_1, ϕ_2 in the induced space, the function $g \mapsto \langle \phi_1(g), \phi_2(g)\rangle$ transforms on the left under $\widetilde{B}$ by δ_B. So its integral over $\widetilde{B}\backslash\widetilde{G}$ is a $\widetilde{G}$-invariant inner product.

(b) By Frobenius reciprocity and Lemma 7.1,

$$\dim \mathrm{End}_{\widetilde{G}}(\mathrm{I}(\chi)) = \dim \mathrm{Hom}_{\widetilde{A}}(r_N\,\mathrm{I}(\chi), \chi) \leq \dim \mathrm{Hom}(\chi,\chi) + \dim \mathrm{Hom}(\chi^{\tilde{w}}, \chi) = 1 + 0.$$

Here $\chi = \chi_\lambda$ and $\chi^{\tilde{w}} \cong \chi_{\lambda^w}$ are irreducible and not isomorphic. Now suppose $W \subset \mathrm{I}(\chi)$ is a nonzero proper invariant subspace. The orthogonal complement $W^\perp$ of W in $\mathrm{I}(\chi)$, for the inner product of (a), is invariant. Each space of $s(K_m)$-fixed vectors is finite-dimensional, by admissibility, and the orthogonal projection onto it is $\Pi(e_m)$ for the normalized genuine idempotent e_m of $s(K_m)$. That projection is self-adjoint and preserves W. Splitting each fixed space orthogonally therefore gives $\mathrm{I}(\chi) = W \oplus W^\perp$. The projection onto W is then a $\widetilde{G}$-endomorphism that is not a scalar, a contradiction. □

Remark 2026.1. *Status: cited only.* The proofs of Lemma 7.1 and Proposition 7.2 use three general facts about ℓ-groups: exactness of coinvariants, $C_c^\infty(N)_N \cong \mathbb{C}$, and Frobenius reciprocity [BZ76, BZ77]. They do not depend on the cover, but their precise theorem numbers were not checked. Luo [Lu20] proves the Knapp–Stein dimension theorem for finite central covers, which sharpens Proposition 7.2(b). The unitary principal series are also tempered. That is standard for linear groups, and [Li12] develops the tempered theory for covers, but it was not checked here.

Lemma 7.3 (Spherical vectors)**.** *Assume* $\gcd(p, 2n) = 1$. *Call* λ unramified *if* $\lambda((h(u), 1)) = 1$ *for all* $u \in (\mathscr{O}^\times)^d$. *Then* $\dim \mathrm{I}(\chi_\lambda)^{s(K)}$ *is* 1 *if* λ *is unramified, and* 0 *otherwise.*

Proof. Since $\widetilde{G} = \widetilde{B}s(K)$, the map $\phi \mapsto \phi(1)$ identifies $\mathrm{I}(\chi)^{s(K)}$ with the vectors in V_χ fixed by $\widetilde{B} \cap s(K) = s(B \cap K)$. The image of $N \cap K$ acts trivially, so these are the vectors fixed by $s(A \cap K) = \{\tilde{h}(u) : u \in \mathscr{O}^\times\}$. This group lies in $\widetilde{C}$, since $A \cap K \subset C$.

The subgroup $\widetilde{C}$ is normal in $\widetilde{A}$, since it contains the center and $\widetilde{A}/Z_A$ is abelian. By Mackey's formula, the restriction of χ_λ to $\widetilde{C}$ is the sum of the d conjugates $\tilde{\lambda}^b$ with $b = \tilde{h}(\varpi^j)$, $0 \le j < d$. By Lemma 5.1, $\tilde{h}(\varpi^j)^{-1}\tilde{h}(u)\tilde{h}(\varpi^j) = (u, \varpi^j)_n^{-2}\tilde{h}(u) = (\varpi, u)_n^{2j}\tilde{h}(u)$, so $\tilde{\lambda}^{\tilde{h}(\varpi^j)}(\tilde{h}(u)) = (\varpi, u)_n^{2j}\,\tilde{\lambda}(\tilde{h}(u))$. On $\mathscr{O}^\times$ the character $u \mapsto (\varpi, u)_n^2$ has order exactly d, by (H5). So the restrictions of the d conjugates to $s(A \cap K)$ run once through the d characters of $\mathscr{O}^\times$ that agree with λ on $(\mathscr{O}^\times)^d$. If λ is unramified, exactly one of them is trivial. Otherwise none is. □

Example 7.4. Take $n = 3$ and $p > 3$. Let λ be unramified and put $\alpha = \lambda(\tilde{h}(\varpi^3))$. Then $\mathrm{I}(\chi_\lambda)$ has a unique spherical line. By (H5), $(\varpi^3, -1)_3 = (\varpi, -1)_3^3 = 1$, so Lemma 5.4(c) gives $\lambda^w(\tilde{h}(\varpi^3)) = \lambda(\tilde{h}(\varpi^{-3})) = \alpha^{-1}$. If λ is unitary and $\alpha \neq \pm 1$, then χ_λ is Weyl regular and $\mathrm{I}(\chi_\lambda)$ is irreducible by Proposition 7.2(b).

7.2 Reducibility

Reducible unitary principal series occur only for odd n. Nothing in this subsection is proved here.

Remark 2026.2. *Status: source checked* (statements and introductions of [GS16] and [Sz25]). Moen [Mo88] proves irreducibility of certain unitary principal series of covers of $\mathrm{SL}(2)$ for odd p. Goldberg and Szpruch [GS16, Thm. 5.2], assuming $\mu_n \subset F$ and $p \nmid n$, show that a genuine principal series induced from unitary data is reducible if and only if n is odd and the projection of the central character of the inducing representation to the dth powers is a nontrivial quadratic character. Szpruch [Sz25, §3.1] explains that these representations are parametrized by the nontrivial quadratic characters of $F^\times$, and that each is a sum of two inequivalent irreducible representations. These constituents must be included in any complete account of the unitary dual. This agrees with claim (c) of Table 1.

7.3 Complementary series and endpoints

A *complementary series* representation is an irreducible unitarizable $\mathrm{I}(\chi)$ with χ not unitary. These representations are not tempered. Fix a unitary genuine character λ_0 of Z_A and consider the family $\lambda_s(\zeta\tilde{h}(u)) = \lambda_0(\zeta\tilde{h}(u))|u|^{s/d}$ for $u \in (F^\times)^d$, so that $\lambda_s(\tilde{h}(t^d))$ behaves like $|t|^s$. An invariant Hermitian form on $\mathrm{I}(\chi_{\lambda_s})$, when it exists, is built from a suitably normalized standard intertwining operator $M(s)$. Its positivity is a substantive statement. It does not follow from the uniqueness of invariant Hermitian forms, which only controls a scalar once irreducibility is known.

Converting the parameter. The 1999 manuscript induces from a character μ of $\widetilde{C}$ of the form $\mu_0(x)|x|^{s_{\mathrm{old}}}$, where x is the torus coordinate of $h(x) \in C$. On the common subgroup Z_A, where $x = t^d$, this gives $|t^d|^{s_{\mathrm{old}}}$, while λ_s gives $|t|^s$. So, formally,

$$s = d\, s_{\mathrm{old}}. \tag{7.3}$$

This is a comparison of exponents on Z_A only. The two papers use different subgroups and sections, so (7.3) must be checked against each source before endpoint values are copied. Under (7.3) the 1999 endpoint $s_{\mathrm{old}} = 1/n$ becomes $s = d/n$, which is 1 for odd n and $1/2$ for even n.

Remark 2026.3. *Status: cited only*, except for Aritürk's reducibility points, which are source checked against his introduction. For the double cover with p odd, Gelbart and Sally [GS75, Ge76] treat the complementary series and the endpoint where the even Weil representation appears. For $n = 2$ we have $d = 1$, so the two parameters agree and the endpoint is $s = 1/2$. Aritürk [Ar80] treats the threefold cover. In his parameter, the genuine principal series off $\mathrm{Re}\, s = 0$ are irreducible except at $s = \pm 1/3$, and he describes the composition series there. The 1999 manuscript asserted, for every n, that $0 < s_{\mathrm{old}} < 1/n$ gives complementary series and that the endpoint $s_{\mathrm{old}} = 1/n$ has a square-integrable subrepresentation and a non-tempered spherical quotient. It proved unitarizability by extending the representation to a cover of $\mathrm{GL}(2, F)$ and citing [FK86, §§26–27], but the extension was not proved. It deduced unitarity of the quotient from [KP84, Thm. II.2.1]. These are claims (d) and (e) of Table 1. They are not proved here, they may need the hypothesis $p \nmid n$, and their endpoints must be converted by (7.3).

7.4 The intertwining-operator approach

A complementary-series theorem for general n would come from the positivity of a normalized intertwining operator. We describe the steps and where the difficulty lies. The standard intertwining operator is

$$(T_s\varphi)(g) = \int_F \varphi\big(\tilde{w}^{-1}(u(x), 1)\, g\big)\, dx.$$

Where the integral converges, it maps $\mathrm{I}(\chi_{\lambda_s})$ to $\mathrm{I}(\chi_{\lambda_s}^{\tilde{w}})$. Following Aritürk [Ar80], one can write its action in a Kirillov-type model as a $d \times d$ matrix J_s. Three separate steps are then needed.

(i) Meromorphic continuation and normalization of T_s, with the correct Weyl-conjugate inducing datum.

(ii) Construction of a nondegenerate invariant Hermitian form from T_s, for a family that is Hermitian self-dual.

(iii) Positivity of that form, equivalently of the normalized matrix J_s, on the whole proposed interval.

None of these follows from the corresponding fact for a cover of $\mathrm{GL}(2,F)$ until one proves that the representations of $\widetilde{G}$ extend. When $d>1$ the matrix J_s is not a scalar, which makes step (iii) harder. The local coefficient matrices of [GS16, Sz19] are a natural tool for steps (i) and (ii). For example, when $n=2$ we have $d=1$, J_s is a scalar function of s, and the classical analysis of [GS75] applies.

7.5 Supercuspidal representations and completeness

Completing the unitary dual needs, besides the principal series, the supercuspidal representations and the unitary subquotients of the reducible induced representations.

Remark 2026.4. *Status: cited only.* An irreducible admissible genuine representation is *supercuspidal* if its Jacquet module along N vanishes. By the analogue of Jacquet's theorem for covers [BZ77, BD84], every irreducible admissible genuine representation is either supercuspidal or a subquotient of some $\mathrm{I}(\chi)$. The supercuspidal representations of the double cover are described in [Ma84a, Ma84b], and those of the n-fold cover with $\gcd(p,2n)=1$ in [Jo99]. These citations do not by themselves give the unitary dual for all n, because one must still decide which subquotients of the $\mathrm{I}(\chi)$ are unitary. For example, for $n=3$ this requires the composition series at $s=\pm 1/3$ from [Ar80].

8 Principal-series characters

The trace of $\mathrm{I}(\chi_\lambda)$ can be computed directly from the induced model, and its character then follows from the Weyl integration formula. The character involves Weyl conjugation in the cover, which is not the same as inversion.

Lemma 8.1 (Trace of an induced representation)**.** *Assume* $\gcd(p,2n)=1$. *Let* χ *be a genuine smooth finite-dimensional representation of* $\widetilde{A}$ *and* $\Pi=\mathrm{I}(\chi)$. *For* $f\in C_c^\infty(\widetilde{G})_{\mathrm{gen}}$, *the operator* $\Pi(f)$ *has finite rank and*

$$\operatorname{tr}\Pi(f)=\int_{\widetilde{A}}\operatorname{tr}\chi(a)\,\Psi_f(a)\,da,$$

with Ψ_f *as in* (6.5).

Proof. The compact model. Since $\widetilde{G}=\widetilde{B}s(K)$, restriction to $s(K)$ identifies the induced space with the space H of locally constant $\phi:K\to V_\chi$ with $\phi(bk)=\chi(s(b))\phi(k)$ for $b\in B\cap K$. We regard H as a subspace of $L^2(K,V_\chi)$, using a fixed inner product on V_χ.

The kernel. For $\phi\in H$ and $k\in K$, use the invariance of dg and (3.4):

$$\Pi(f)\phi(k)=\int_{\widetilde{G}}f(s(k)^{-1}g)\phi(g)\,dg=\int_K\mathscr{K}(k,k')\phi(k')\,dk',$$

where $\mathscr{K}(k,k')=\int_{\widetilde{B}}f(s(k)^{-1}bs(k'))\delta_B(b)^{1/2}\chi(b)\,db$ and $db=da\,du$. Choose m such that f is left and right invariant under $s(K_m)$, up to the genuine factor. Then $\mathscr{K}(k,k')$ depends only on the cosets kK_m and $k'K_m$. So $\mathscr{K}$ is a finite sum $\sum_{i,j}1_{k_iK_m}(k)\,1_{k_jK_m}(k')\,M_{ij}$ with $M_{ij}\in\mathrm{End}(V_\chi)$.

The operator on L^2. Define T on all of $L^2(K,V_\chi)$ by $T\phi(k) = \int_K \mathscr{K}(k,k')\phi(k')\,dk'$. By the previous paragraph, T has finite rank, its image consists of locally constant functions, and $\operatorname{tr} T = \int_K \operatorname{tr}\mathscr{K}(k,k)\,dk$. A substitution $b \mapsto s(b_0)b$ shows that $\mathscr{K}(b_0k,k') = \chi(s(b_0))\mathscr{K}(k,k')$ for $b_0 \in B\cap K$. So the image of T lies in H. On H, T agrees with $\Pi(f)$.

The trace. Let E be the image of T, a finite-dimensional subspace of H. Both T, on $L^2(K,V_\chi)$, and $\Pi(f)$, on H, map into E, and they agree on E. The trace of a finite-rank operator equals the trace of its restriction to any finite-dimensional invariant subspace that contains its image. So $\operatorname{tr}\Pi(f) = \operatorname{tr}(T|_E) = \operatorname{tr} T$. Hence

$$\operatorname{tr}\Pi(f) = \int_K \operatorname{tr}\mathscr{K}(k,k)\,dk = \int_{\widetilde{A}}\int_N \operatorname{tr}\chi(a)\,\delta_B(a)^{1/2} f^K(a(u,1))\,du\,da,$$

because χ is trivial on the image of N. This is the stated formula. □

Corollary 8.2 (Trace and orbital integrals)**.** *Assume* $\gcd(p,2n) = 1$. *Let λ be a genuine smooth character of Z_A and $\Pi = \mathrm{I}(\chi_\lambda)$. For $f \in C_c^\infty(\widetilde{G})_{\mathrm{gen}}$,*

$$\operatorname{tr}\Pi(f) = \frac{1}{2}\int_{Z_A} \big(\operatorname{tr}\chi_\lambda(z) + \operatorname{tr}\chi_\lambda^{\tilde{w}}(z)\big)\Phi_f(z)\,dz \tag{8.1}$$

$$= d\int_{Z_A} \lambda(z)\Phi_f(z)\,dz. \tag{8.2}$$

Proof. By Theorem 5.6, $\operatorname{tr}\chi_\lambda$ vanishes off Z_A and equals $d\lambda$ on Z_A. Lemma 8.1 and Lemma 6.3 give (8.2), since the singular set has measure zero. The integrand is invariant under μ_n, because λ and Φ_f transform by inverse characters. The map $z \mapsto \tilde{w}^{-1}z\tilde{w}$ preserves the measure on Z_A, since it is $\zeta\tilde{h}(u) \mapsto \zeta\tilde{h}(u^{-1})$. Together with the Weyl invariance of Φ_f, this shows that the two terms of (8.1) are equal. So (8.1) equals (8.2). □

Theorem 8.3 (Principal-series character)**.** *Assume* $\gcd(p,2n) = 1$. *Let λ be a genuine smooth character of Z_A, $\chi = \chi_\lambda$ and $\Pi = \mathrm{I}(\chi)$, which need not be irreducible. Define Θ_Π on the regular semisimple set of $\widetilde{G}$ by*

$$\Theta_\Pi(x^{-1}ax) = \frac{\operatorname{tr}\chi(a) + \operatorname{tr}\chi^{\tilde{w}}(a)}{|D(a)|^{1/2}} \qquad (a \in \widetilde{A}_{\mathrm{reg}},\ x \in \widetilde{G}), \tag{8.3}$$

and $\Theta_\Pi = 0$ on elliptic regular elements. Then Θ_Π is a well-defined, locally integrable class function, and $\operatorname{tr}\Pi(f) = \int_{\widetilde{G}}\Theta_\Pi(g)f(g)\,dg$ for all $f \in C_c^\infty(\widetilde{G})_{\mathrm{gen}}$. In particular $\Theta_\Pi = 0$ on $\widetilde{A}_{\mathrm{reg}} \setminus Z_A$, and for $z \in (Z_A)_{\mathrm{reg}}$

$$\Theta_\Pi(z) = \frac{d\big(\lambda(z) + \lambda(\tilde{w}^{-1}z\tilde{w})\big)}{|D(z)|^{1/2}}. \tag{8.4}$$

It satisfies $\Theta_\Pi(\zeta g) = \zeta\Theta_\Pi(g)$.

Proof. Well defined. The right side of (8.3) is invariant under conjugation of a by $\widetilde{A}$, since traces are. It is also invariant under $a \mapsto \tilde{w}^{-1}a\tilde{w}$, since $\operatorname{tr}\chi^{\tilde{w}}(\tilde{w}^{-1}a\tilde{w}) = \operatorname{tr}\chi(\tilde{w}^{-2}a\tilde{w}^2) = \operatorname{tr}\chi(a)$ by Lemma 5.4(d). If $x^{-1}ax = y^{-1}a'y$ with $a,a' \in \widetilde{A}_{\mathrm{reg}}$, then yx^{-1} conjugates a to a' and lies in $\widetilde{A}\cup\tilde{w}\widetilde{A}$. So the value does not depend on the choices. Formula (8.4) and the vanishing off Z_A follow from (5.5), because $\tilde{w}^{-1}a\tilde{w} \in Z_A$ exactly when $a \in Z_A$.

The trace. Let $f \in C_c^\infty(\widetilde{G})_{\mathrm{gen}}$. The singular semisimple set has Haar measure zero, so it suffices to integrate over regular elements, where Θ_Π is defined. The product $\Theta_\Pi f$ is invariant under μ_n, so it is the pullback of a function $\bar\phi$ on G, and $\int_{\widetilde{G}} \Theta_\Pi f = \int_G \bar\phi$ because μ_n has mass one. Apply (3.9). The elliptic term vanishes because Θ_Π does. For $a = h(t)$ with $t \notin (F^\times)^d$ the integrand vanishes. For $t \in (F^\times)^d$ let $z \in Z_A$ lie over $h(t)$. Then $\bar\phi(x^{-1}h(t)x) = \Theta_\Pi(z) f(\tilde x^{-1} z \tilde x)$ for any lift $\tilde x$ of x, and $A\backslash G$ has the same measure as $\widetilde{A}\backslash\widetilde{G}$. So

$$\int_{\widetilde{G}} \Theta_\Pi f = \frac{1}{2}\int_{Z_A} |D(z)|\Theta_\Pi(z) \int_{\widetilde{A}\backslash\widetilde{G}} f(x^{-1}zx)\,d\dot x\,dz = \frac{1}{2}\int_{Z_A} \big(\operatorname{tr}\chi(z) + \operatorname{tr}\chi^{\tilde w}(z)\big)\Phi_f(z)\,dz,$$

where we used that the integrand is invariant under μ_n. By Corollary 8.2 this is $\operatorname{tr}\Pi(f)$.

Local integrability. Let $\Omega \subset \widetilde{G}$ be compact. We may assume that Ω is open and stable under μ_n, so that $\omega = \rho(\Omega)$ is compact open and 1_Ω is the pullback of $1_\omega \in C_c^\infty(G)$. The function $|\Theta_\Pi|$ is invariant under μ_n. The same computation as above, with $|\Theta_\Pi| 1_\Omega$ in place of $\Theta_\Pi f$, gives

$$\int_\Omega |\Theta_\Pi| = \frac{1}{2}\int_{(F^\times)^d} \big|\operatorname{tr}\chi(z_t) + \operatorname{tr}\chi^{\tilde w}(z_t)\big|\, F^A_{1_\omega}(t)\,dt,$$

where $z_t \in Z_A$ is any lift of $h(t)$. By (3.8), $F^A_{1_\omega}$ is bounded and compactly supported. The traces are continuous, so they are bounded on the lifts of that support. Hence the integral is finite.

Equivariance. Both $\chi(\zeta a)$ and $\chi^{\tilde w}(\zeta a)$ are ζ times their values at a. □

Remark 8.4. Θ_Π is the character of the induced representation, which may be reducible. When $\mathrm{I}(\chi_\lambda)$ is reducible, Θ_Π is the sum of the characters of its constituents. The paper does not compute the characters of the individual constituents.

Caution 8.5 (Inversion is not Weyl conjugation). For $n > 2$ the expression $\operatorname{tr}\chi(a) + \operatorname{tr}\chi(a^{-1})$ transforms under $a \mapsto \zeta a$ by ζ in the first term and by ζ^{-1} in the second. Its quotient by $|D(a)|^{1/2}$ is therefore not the character of a genuine representation, and it cannot be used in a matching theorem. By Lemma 5.4(b), Weyl conjugation looks like inversion on the matrix part. It differs on the central part, and Example 5.5 shows that it can differ even on $\tilde h(t)$.

Example 8.6. Take $n = 3$ and $p > 3$, and let λ be a genuine character of Z_A with $\alpha = \lambda(\tilde h(\varpi^3))$. As in Example 7.4, $\lambda(\tilde w^{-1}\tilde h(\varpi^3)\tilde w) = \lambda(\tilde h(\varpi^{-3})) = \alpha^{-1}$. Since $|\varpi^3 - \varpi^{-3}| = q^3$, Theorem 8.3 gives

$$\Theta_\Pi(\tilde h(\varpi^3)) = \frac{3(\alpha + \alpha^{-1})}{q^3}, \qquad \Theta_\Pi(\tilde h(\varpi)) = 0.$$

The second value vanishes because $\varpi \notin (F^\times)^3$.

9 A restricted and normalized split-torus transfer

Genuine principal series of $\widetilde{G}$ can be compared with principal series of G through a power map from A to Z_A. That map is a homomorphism only for some n. In this section assume $\gcd(p, 2n) = 1$ and

$$n \text{ is odd} \qquad \text{or} \qquad 4 \mid n. \tag{9.1}$$

Then $n \mid d^2$. Indeed $d = n$ when n is odd, and $d = n/2$ is even when $4 \mid n$. For $t \in F^\times$ define

$$\nu(h(t)) = (h(t^d), 1) \in Z_A. \tag{9.2}$$

Lemma 9.1. *Under* (9.1), $\nu : A \to Z_A$ *is a continuous homomorphism with kernel* $h(\mu_d)$*, of order* d*. Its image is* Z_A^0*, which is a subgroup, and* $Z_A = \mu_n \times Z_A^0$*. Moreover*

$$\tilde{w}^{-1}\nu(h(t))\tilde{w} = \nu(h(t^{-1})). \tag{9.3}$$

If dz *is the restriction to* Z_A^0 *of the measure on* $\widetilde{A}$*, then*

$$\int_{Z_A^0} H(z)\,dz = \frac{1}{d}\int_A H(\nu(h(t)))\,dt \tag{9.4}$$

for every compactly supported locally constant H *on* Z_A^0*.*

Proof. By Lemma 3.2(a), $\beta(h(s^d),h(t^d)) = (t^d,s^d)_n = (t,s)_n^{d^2} = 1$, since $n \mid d^2$. So ν is a homomorphism, its image Z_A^0 is a subgroup, and $Z_A^0 \cap \mu_n = 1$. The kernel is $\{h(t) : t^d = 1\}$, which has order d because $d \mid n$ and $\mu_n \subset F$. Equation (9.3) is Lemma 5.4(b) with t^d in place of t.

For (9.4), the dth power map sends $\varpi^k\mathscr{O}^\times$ onto $\varpi^{dk}(\mathscr{O}^\times)^d$, and every element of $(F^\times)^d$ lies in exactly one of these sets. The first set has volume 1, and the second has restricted volume $1/d$ by Example 3.6. So the pushforward of dt under $t \mapsto t^d$ is d times the restricted measure on $(F^\times)^d$. This is (9.4). □

Definition 9.2 (Matching on the split torus)**.** Let $\bar{f} \in C_c^\infty(\widetilde{G})_{\mathrm{gen}}$ and $f \in C_c^\infty(G)$. They are *split-matching* if

$$F_f^A(t) = \Phi_{\bar{f}}(\nu(h(t))) \qquad (t \in F^\times), \tag{9.5}$$

where both sides are the locally constant extensions of Sections 3.5 and 6. This notion concerns split orbital integrals only.

Example 9.3. The genuine unit e_K and 1_K are split-matching. Indeed $\Phi_{e_K}(\nu(h(t))) = 1_{\mathscr{O}^\times}(t^d) = 1_{\mathscr{O}^\times}(t)$ by Example 6.6, and $F_{1_K}^A(t) = 1_{\mathscr{O}^\times}(t)$ by Example 3.7.

Theorem 9.4 (Restricted split-torus trace matching)**.** *Assume* $\gcd(p,2n) = 1$ *and* (9.1)*. For a genuine smooth character* λ *of* Z_A *put*

$$\eta_\lambda(t) = \lambda(\nu(h(t))), \qquad t \in F^\times. \tag{9.6}$$

This is a smooth character of $A \cong F^\times$*, unitary if* λ *is. If* $\bar{f}$ *and* f *are split-matching, then*

$$\boxed{\mathrm{tr}\,\mathrm{I}(\chi_\lambda)(\bar{f}) = \mathrm{tr}\,\mathrm{Ind}_B^G(\eta_\lambda\delta_B^{1/2})(f)} \tag{9.7}$$

Every $\bar{f} \in C_c^\infty(\widetilde{G})_{\mathrm{gen}}$ *has at least one split-matching* $f \in C_c^\infty(G)$*. Neither induced representation needs to be irreducible.*

Theorem 9.4 is an equality of traces for split-matching test functions, and nothing more. It does not give a transfer operator, a matching of elliptic orbital integrals, a correspondence between all irreducible representations, or identities for the two constituents of a reducible principal series.

Proof. By Lemma 9.1, η_λ is a character. The integrand $\lambda\Phi_{\bar f}$ is invariant under μ_n, and μ_n has mass one. By Corollary 8.2 and Lemma 9.1,

$$\operatorname{tr} \mathrm{I}(\chi_\lambda)(\bar f) = d\int_{Z_A^0} \lambda(z)\Phi_{\bar f}(z)\,dz = \int_A \eta_\lambda(t)\Phi_{\bar f}(\nu(h(t)))\,dt = \int_A \eta_\lambda(t) F_f^A(t)\,dt.$$

By (3.11) the last integral is $\operatorname{tr}\operatorname{Ind}_B^G(\eta_\lambda \delta_B^{1/2})(f)$.

For existence, $\Phi_{\bar f} \in C_c^\infty(Z_A)^W_{\mathrm{gen}}$ by Theorem 6.4. The map ν is proper, since its kernel is finite and its image is closed. So $t \mapsto \Phi_{\bar f}(\nu(h(t)))$ is locally constant and compactly supported, and it is invariant under $t \mapsto t^{-1}$ by (9.3). The case $n = 1$ of Theorem 6.4 gives $f \in C_c^\infty(G)$ with this function as F_f^A. □

Example 9.5 (The normalization in degree three)**.** Take $n = 3$ and $p > 3$, and let λ be unramified. By Example 6.6 and Example 3.6,

$$\operatorname{tr} \mathrm{I}(\chi_\lambda)(e_K) = 3\int_{Z_A} \lambda(z)\Phi_{e_K}(z)\,dz = 3\,\mathrm{vol}\big((\mathscr{O}^\times)^3\big) = 3\cdot\tfrac{1}{3} = 1.$$

This agrees with Lemma 7.3 and Example 3.5. It also agrees with Theorem 9.4, since e_K matches 1_K (Example 9.3) and the right side is 1 by Example 3.8. The example shows why the factor in (8.2) is d. Using the index d^2 there would give 3 instead of 1.

Remark 9.6 (The case $n \equiv 2 \pmod 4$). If $n \equiv 2 \pmod 4$, then d is odd and $n \nmid d^2$. The map $t \mapsto (h(t^d), 1)$ is then not a homomorphism in general, and Z_A^0 is not a subgroup. The phrase "twist by the Weil index" does not by itself define a μ_n-valued homomorphism here. Weil indices take values in μ_4 and need a compatible normalization or a larger central group. So (9.7) is not claimed for $n \equiv 2 \pmod 4$, and in particular not for $n = 2$. This is a limitation of the particular power map ν, not a statement that no transfer exists in these degrees. The double cover must be treated with its own genuine characters and transfer factors [Ge76, GS75, Fl80, Sc98].

Remark 9.7 (Spherical functions). The identity (9.7) holds whenever $\bar f$ and f are spherical and split-matching. It is not proved here that every spherical $\bar f$ has a spherical partner. That is a statement about the metaplectic Satake transform. Example 9.3 shows that it holds for the units.

10 Summary of results and open questions

We collect what is proved, what is cited, a running example, and the open questions.

Proved here. Lemma 5.2 and Theorem 5.6 describe the covering torus. They distinguish three integers that are easily confused:

$$\dim \chi_\lambda = d, \qquad [\widetilde{A} : Z_A] = d^2, \qquad |\ker \nu| = d.$$

Lemma 6.1 and Theorem 6.4 give the vanishing and the image of split orbital integrals. Lemma 8.1, Corollary 8.2 and Theorem 8.3 give the trace and the character of genuine principal series. Proposition 7.2 and Lemma 7.3 give unitarity, irreducibility for Weyl regular data and the spherical vectors. Theorem 9.4 gives the split-torus character identity for $\gcd(p, 2n) = 1$ and n odd or divisible by 4. For instance, for $n = 3$ and $F = \mathbb{Q}_7$ it compares $\mathrm{I}(\chi_\lambda)$ with $\operatorname{Ind}_B^G(\eta_\lambda \delta_B^{1/2})$, where $\eta_\lambda(t) = \lambda(\tilde h(t^3))$.

Not proved here. Remarks 2026.1 (cited only), 2026.2 (source checked), 2026.3 (mostly cited only) and 2026.4 (cited only). Claims (d), (e) and (f) of Table 1 are neither proved nor refuted.

A running example. Table 2 collects the threefold cover over $F = \mathbb{Q}_7$, with $\varpi = 7$ and $q = 7$. Here $d = 3$.

Quantity	Value	Where
$(7,3)_3$	image of 4 mod 7, a primitive cube root of 1	Example 3.1
$\tilde{h}(7)\tilde{h}(3)$ versus $\tilde{h}(3)\tilde{h}(7)$	differ by a nontrivial cube root of 1	Example 3.4
$[\widetilde{A} : Z_A]$, $[\widetilde{A} : \widetilde{C}]$, $\dim \chi_\lambda$	9, 3, 3	Lemma 5.2, Example 5.7
$\mathscr{O}_f^{Z_A}(\tilde{h}(7))$	0, since $7 \notin (\mathbb{Q}_7^\times)^3$	Example 6.2
$\mathrm{vol}\big((\mathbb{Z}_7^\times)^3\big)$	$1/3$	Example 3.6
$\Phi_{e_K}(\zeta\tilde{h}(u))$, $u \in (\mathbb{Q}_7^\times)^3$	$\zeta^{-1} 1_{\mathbb{Z}_7^\times}(u)$	Example 6.6
$\dim \mathrm{I}(\chi_\lambda)^{s(K)}$, λ unramified	1	Lemma 7.3
$\Theta_\Pi(\tilde{h}(7^3))$, $\Theta_\Pi(\tilde{h}(7))$	$3(\alpha+\alpha^{-1})/7^3$, 0	Example 8.6
$\mathrm{tr}\,\mathrm{I}(\chi_\lambda)(e_K)$	$1 = \mathrm{tr}\,\mathrm{Ind}_B^G(\eta_\lambda \delta_B^{1/2})(1_K)$	Example 9.5
$\eta_\lambda(t)$	$\lambda(\tilde{h}(t^3))$	Theorem 9.4

Table 2: The threefold cover of $\mathrm{SL}(2,\mathbb{Q}_7)$. Here $\alpha = \lambda(\tilde{h}(7^3))$.

Open questions. These are logically independent.

(1) *The 1999 and 2001 unitary-dual claims.* For $\gcd(p,2n) = 1$, independently prove or refute claims (d)–(f) of Table 1: the complementary intervals, the constituents at their endpoints, and the unitarity of those constituents.

(2) *Wild residue characteristic.* Treat $p \mid n$. The center is still Z_A by Lemma 5.2, but the index $[\widetilde{A} : Z_A]$ and the splitting of K change. The work of Szpruch [Sz19, Sz25] does not assume $p \nmid n$.

(3) *Degrees $n \equiv 2 \pmod 4$.* The power map ν fails here. The map $t \mapsto (h(t^{2d}), 1)$ is a homomorphism, since $(t,s)_n^{4d^2} = 1$, but its image is smaller than Z_A^0. Find a transfer in these degrees, possibly with a Weil-index normalization.

(4) *Spherical matching.* Decide whether every spherical genuine function has a spherical split-matching partner, and relate this to the Satake isomorphism of [Mc12].

(5) *Elliptic matching and constituents.* Extend the comparison to elliptic orbital integrals, and find character identities for the constituents of reducible principal series. For $n = 2$ this was the goal of the 1999 manuscript, as a step toward Labesse's multiplicity one conjecture.

11 Glossary

Alternating pairing

A pairing $\langle a,b\rangle$ with $\langle a,a\rangle = 1$. The commutator on $\widetilde{A}/Z_A$ is one.

Center of $\widetilde{A}$ The group $Z_A = \rho^{-1}(h((F^\times)^d))$. It is the radical of the commutator pairing.

Complementary series
Irreducible unitarizable principal series induced from a nonunitary character. They are not tempered.

Genuine A representation on which $\zeta \in \mu_n$ acts by ζ. A test function with $f(\zeta g) = \zeta^{-1} f(g)$.

Genuine unit The function e_K with $e_K(\zeta s(k)) = \zeta^{-1}$. It acts as the projection onto spherical vectors.

Jacquet module The coinvariants of a representation under N, twisted by $\delta_B^{-1/2}$. For $\mathrm{I}(\chi)$ it has constituents χ and $\chi^{\tilde{w}}$.

Hilbert symbol The pairing $(\ ,\ )_n : F^\times \times F^\times \to \mu_n$ of local class field theory.

Kubota cocycle The two-cocycle (3.1) that defines $\widetilde{G}$.

Normalized orbital integral
An orbital integral multiplied by $|D|^{1/2}$. Its quotient group must centralize the element.

Normalized induction
Induction from $\chi \otimes \delta_B^{1/2}$. It preserves unitarity.

Power map The map $\nu(h(t)) = (h(t^d), 1)$. It is a homomorphism when n is odd or $4 \mid n$.

Principal series A representation $\mathrm{I}(\chi)$ induced from a genuine irreducible representation of $\widetilde{A}$.

Spherical Fixed by $s(K)$, for a fixed splitting s.

Split-matching Equality (9.5) of split orbital integrals. It is not a stable or endoscopic transfer.

Stone–von Neumann theorem
The parametrization of irreducible representations of a two-step nilpotent group by central characters.

Supercuspidal An irreducible admissible representation whose Jacquet module along N vanishes.

Tame The case $p \nmid n$, where the Hilbert symbol is given by (H5).

Weyl conjugate The representation $\chi^{\tilde{w}}(a) = \chi(\tilde{w}^{-1} a \tilde{w})$. It is not $\chi(a^{-1})$.

Weyl discriminant
$D(h(t)) = -(t - t^{-1})^2$.

Weyl regular χ_λ with $\lambda^w \neq \lambda$.

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